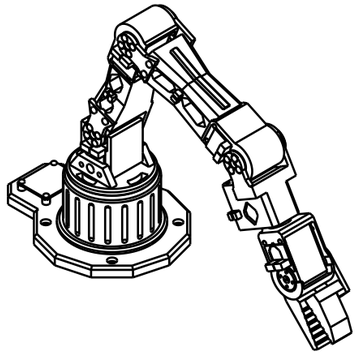
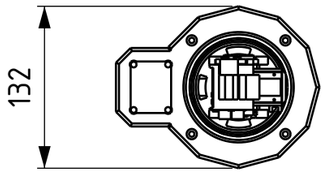
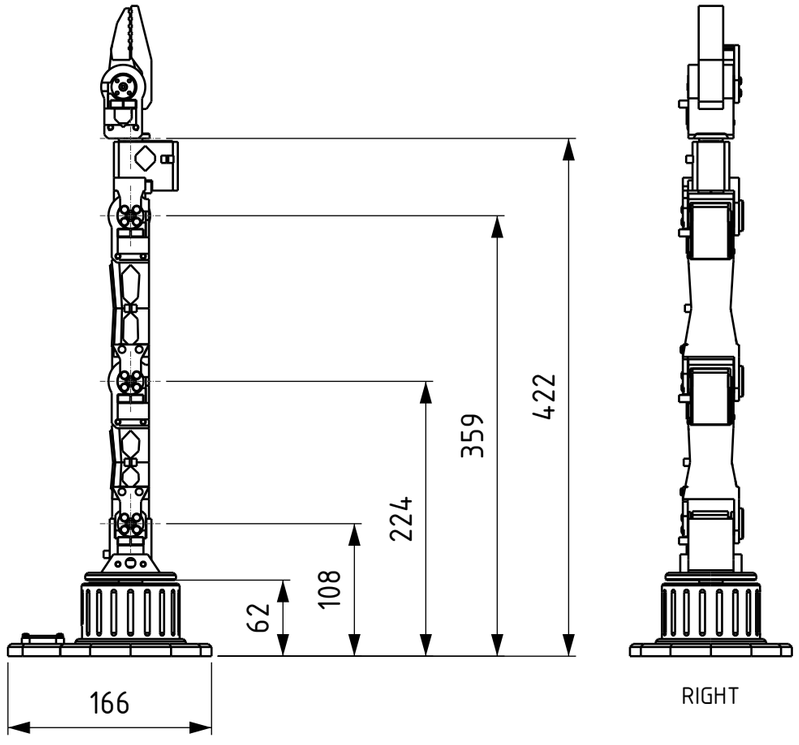


ISOMETRIC (NOT TO SCALE)



TOP



FRONT

RIGHT

PRINTED PARTS

PART NO.	NAME	QTY	SET
TAL-101	Base	1	A
TAL-102	Shoulder (turntable + cradle)	1	A
TAL-103	Upper arm	1	A
TAL-104	Forearm	1	B
TAL-105	Wrist	1	B
TAL-106	Gripper body	1	B
TAL-107	Moving jaw	1	B
TAL-108	Hub cap	3	A
TAL-110	Idler plate	2	B
TAL-111	Servo fit gauge (print first)	1	A

BOUGHT PARTS

ITEM	QTY
Feetech STS3215 servo (12 V: C018 / C047)	6
Waveshare Bus Servo Adapter (A)	1
M3 x 6 button head (output horns)	24
M3 x 5 button head (passive horns)	12
M3 x 10 button head (idler plates)	4
M2 x 4 self-tapping (servo cases)	6
M2 x 5 self-tapping (servo cases)	6
M2 x 6 self-tapping (servo cases)	8
M2.5 x 6 self-tapping (board)	4
M4 bolts to the table (user's)	4
3-pin bus servo cables (with servos)	6

JOINT LAYOUT

1. Pan axis vertical through the base centre; pan horn face 62 above the table. Plinth 132 across corners.
2. Shoulder axis (along Y) 46 above the pan horn face. Upper arm 116, forearm 135 (axis to axis). Wrist flex axis to roll horn face 63.
3. Front view: zero pose (all joints 0). Isometric: 'ready' pose.
4. Assembly, each joint: take both horns off the servo (one centre screw each; the base and wrist roll servos may keep theirs); plug a 3-pin cable into the servo; push the servo into its part along its entry path (part sheet notes); fit its M2 case screws; refit both horns through the R10.8 openings, each with its centre screw; fit the next link with 4X M3 × 6 into the output horn and 4X M3 × 5 into the passive horn.
5. Print TAL-111 first and check the servo fit before printing the arm. Each part sheet gives the print orientation; print-ready 3MF files are in cad/print.

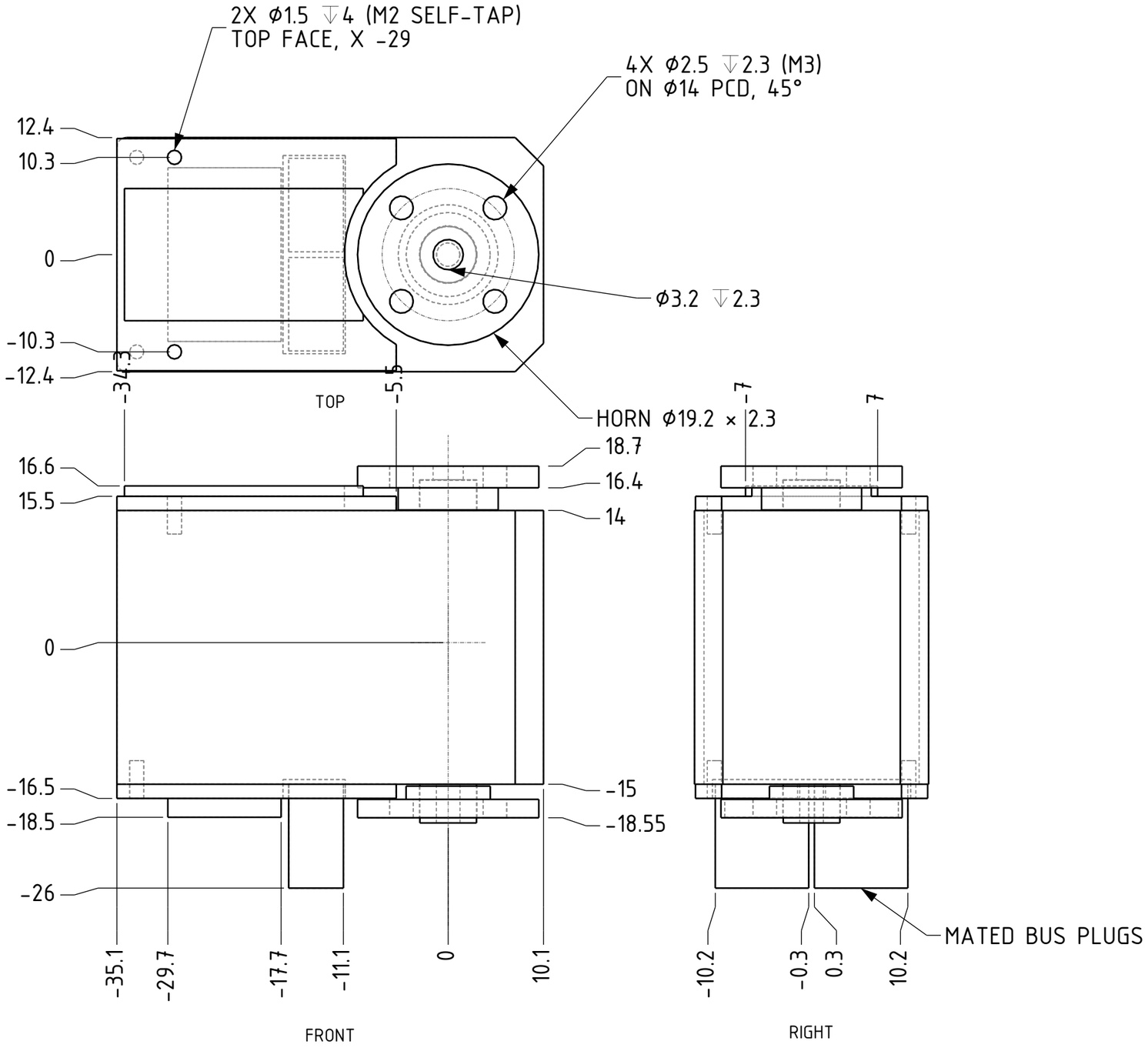
NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part sheets give each part in its own frame; this sheet shows where those frames sit.

THIRD ANGLE PROJECTION

REFERENCE SHEET: in work sets A and B

PROJECT TALOS 5-axis arm	EST. MASS -
PART General assembly and parts list	SCALE 1:5
PART NO. TAL-000	UNITS / REV mm / A
QTY / ARM -	DATE 2026-11-09
MATERIAL -	SHEET 1 of 21



SERVO (servo frame)

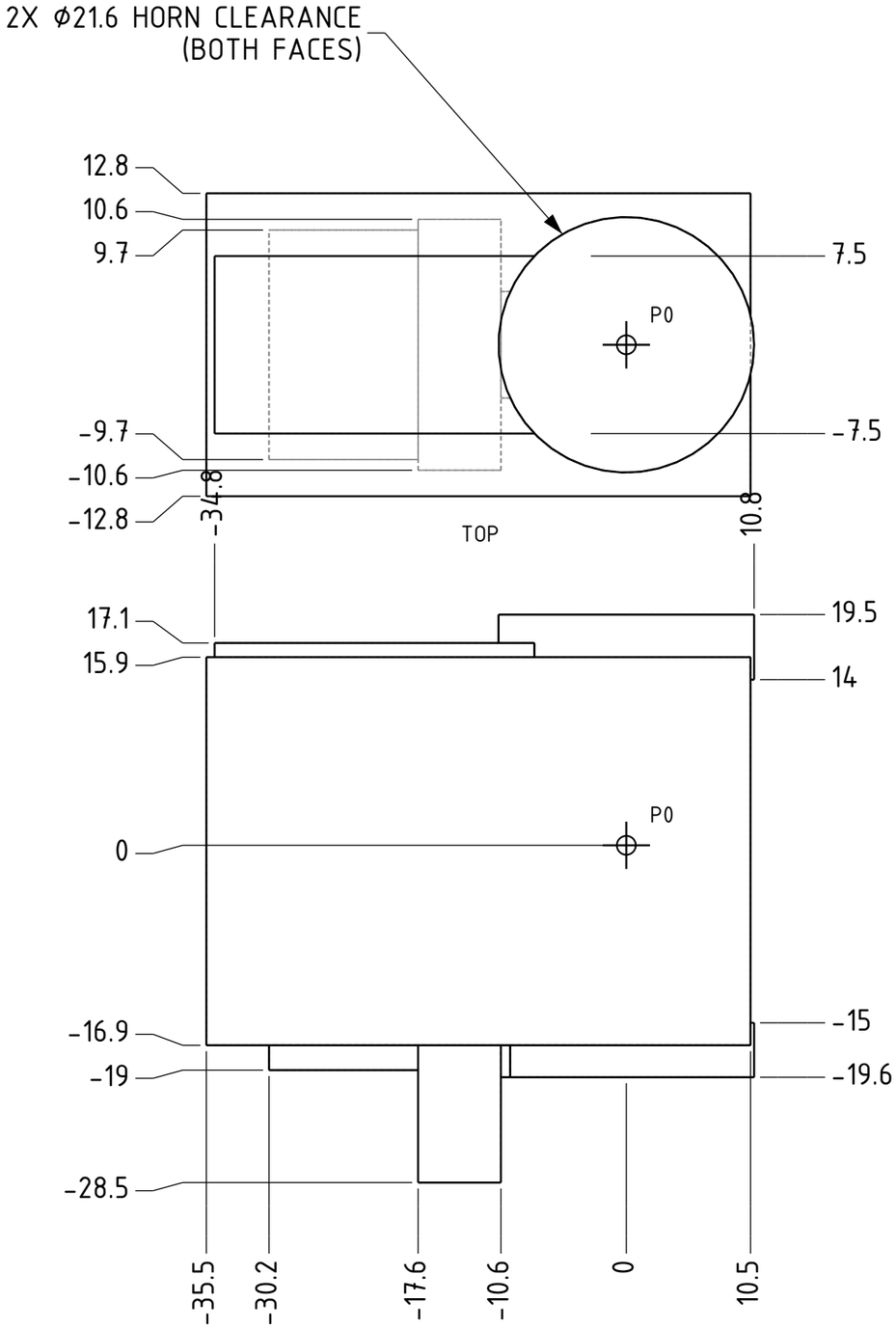
1. Case 45.2 × 24.8, x −35.1 to 10.1, full outline z −15 to 14; +X end corners C3, −X end corners R1.
2. Top face (output side) z 15.5, from x −35.1 to −5.5, stopping at R11 round the shaft. Rib on it: x −34.3 to −9, y ±7, up to z 16.6, its shaft end at R11.
3. Bottom face (connector side) z −16.5, from x −35.1 to −5.5, stopping at R11. Bump on it: x −29.7 to −17.7, y ±9.2, down to z −18.5.
4. Connector sockets x −16.5 to −11.5, y 0.5 to 10 and −10 to −0.5; the plugs go in from the bottom face. Mated plugs modelled to z −26.
5. Output horn ⌀19.2: disc z 16.4 to 18.7, hub ⌀10.6 down to the case (z 14); 4X ⌀2.5 M3 on ⌀14 PCD at 45°, centre ⌀3.2. Bare spline (horn off) ⌀6 up to z 17.2.
6. Passive horn ⌀19.2: disc z −16.6 to −18.55, turning on the idler boss ⌀6 (tip z −19.1).
7. Case holes ⌀1.5 ↧4 (M2 self-tap): top face at (−29, ±10.3), bottom face at (−33, ±10.3).

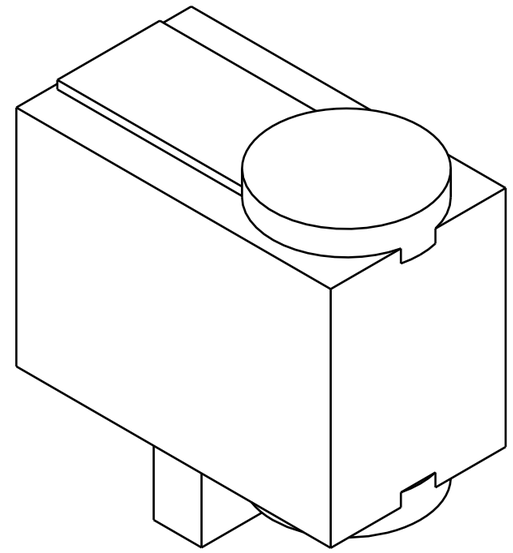
NOTES

- Bought part: Feetech STS3215 (the 7.4 V and 12 V variants share this case). Sizes measured from Feetech's CAD inside SO-101's assembly STEP (cad/reference/so101/step/STS3215_from_SO101.step); each modelled size is the real one or up to 0.25 bigger. servo_reference.py checks every pocket against that model.
- Both horns come off (one centre screw each): every servo goes into its part with its horns off. Sheet 2 gives pocket P, the envelope every servo part cuts.

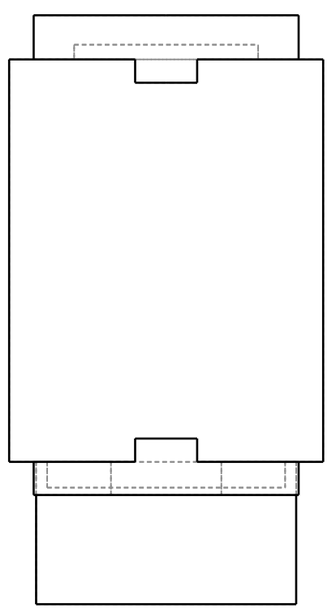
THIRD ANGLE PROJECTION

REFERENCE SHEET: in work sets A and B	
PROJECT TALOS 5-axis arm	EST. MASS -
PART STS3215 servo (reference) (part sheet 1 of 2)	SCALE 2:1
PART NO. TAL-001	UNITS / REV mm / A
QTY / ARM 6	DATE 2026-11-09
MATERIAL bought part	SHEET 2 of 21





ISOMETRIC (NOT TO SCALE)



RIGHT

POCKET P (servo frame; 0.4 / 0.5 clearance included)

1. Case box x −35.5 to 10.5, y ±12.8, z −16.9 to 15.9: the case + 0.4 all round (printing tolerance).
2. Rib channel x −34.8 to −3.5 (into the spline channel), y ±7.5, z 15.5 to 17.1; bump channel x −30.2 to −17.2, y ±9.7, z −19 to −16.5 (each + 0.5).
3. Plug bay box x −17.6 to −10.6, y ±10.6, z −28.5 to −15 (mated plugs + cable bend).
4. Relief box x −10.6 to −9.0, y ±4.5, z −19.6 to −15: no knife edge where the plug bay meets the idler ⌀21.6.
5. Idler cylinder ⌀21.6 on the shaft axis, z −19.6 to −15 (passive horn and boss); horn cylinder ⌀21.6, z 14 to 19.5 (output horn; bare spline ⌀7 to z 17.6).

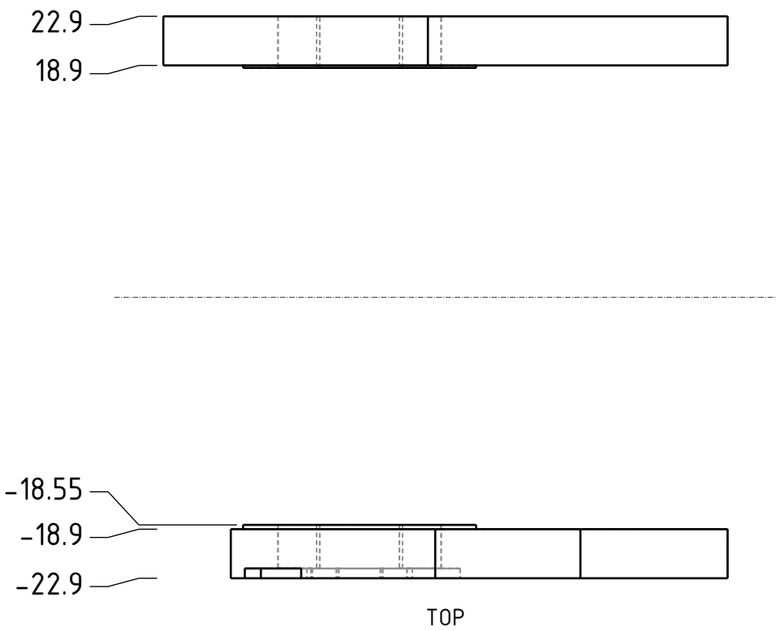
   Way in (each part sheet): the case box, rib, bump and plug bay boxes run on out along the servo's entry direction, and the spline and boss ⌀7 become slots of their width. The servo goes in with both horns off.

NOTES

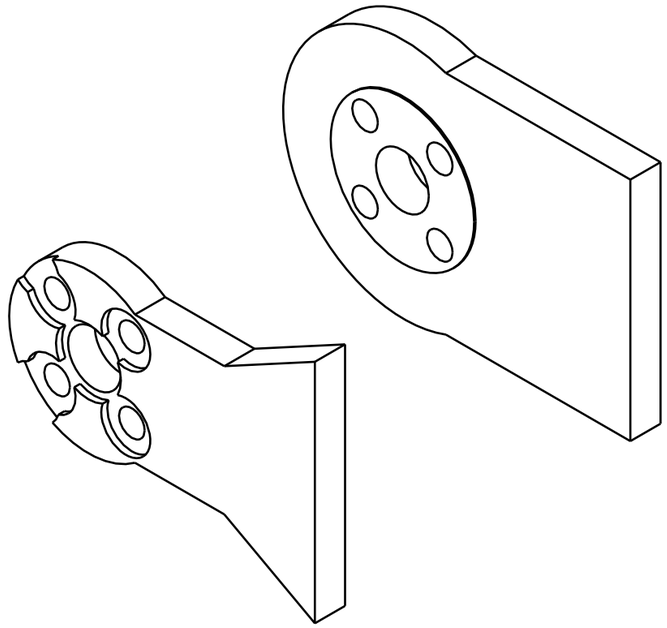
- Each part sheet locates P0 and gives the servo axes in the part frame; cut the union of the bodies.
- All dimensions mm. Third-angle projection. Do not scale the drawing.

THIRD ANGLE PROJECTION

REFERENCE SHEET: in work sets A and B	
PROJECT TALOS 5-axis arm	EST. MASS -
PART Servo pocket P (cut into parts) (part sheet 2 of 2)	SCALE 2:1
PART NO. TAL-001	UNITS / REV mm / A
QTY / ARM -	DATE 2026-11-09
MATERIAL -	SHEET 3 of 21



TOP



ISOMETRIC (NOT TO SCALE)

IDLER NECK POINTS (X, Z), MIRRORED IN Z

PT	X	Z
N1	0	±10.5
N2	6	±8.5
N3	0.6R = 18	±8.5
N4	R = 30	±H = ±15

CLEVIS (every pitch joint)

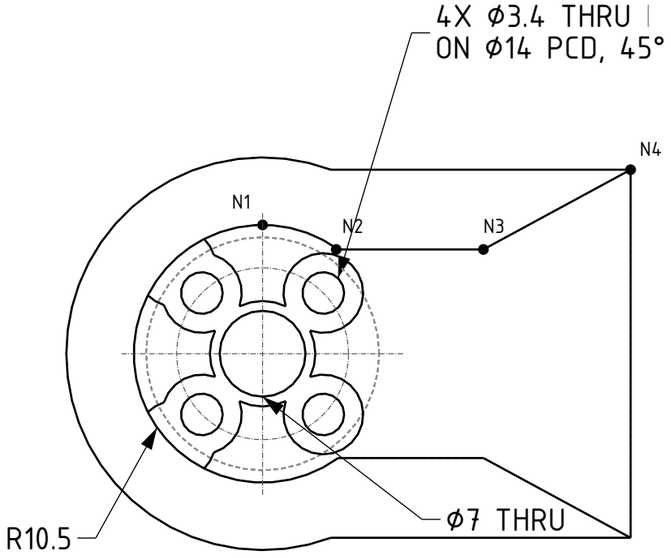
Drawn for the arm links: reach R = 30, half-height H = 15. The wrist uses R = 24, H = 12.

Not printed on its own: the plates are modelled into the links. On TAL-103 and TAL-104 the idler plate is the separate part TAL-110; on the wrist (TAL-105) both plates are wrist body.

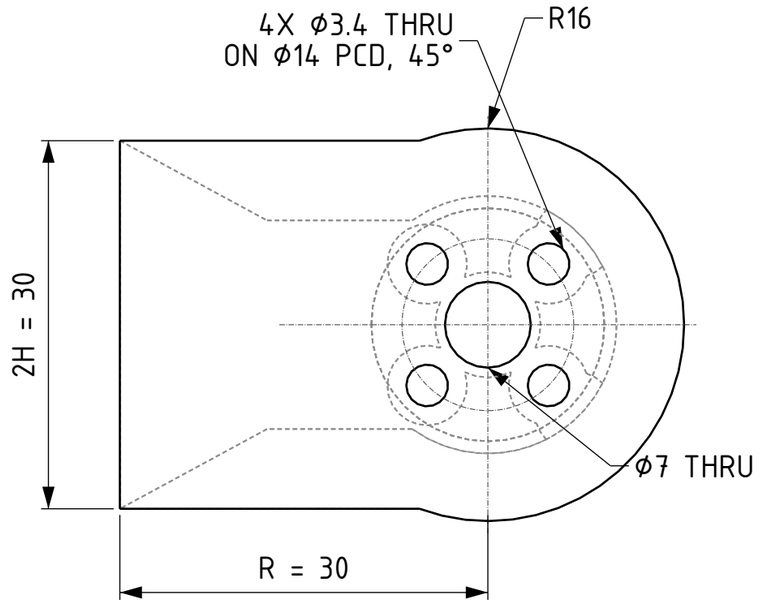
1. Output plate (+Y), y 18.9 to 22.9: circle R16 at the origin + rectangle x 0 to R, z ±H.
2. Output plate pad ⌀19 on the inner face, y 18.7 to 18.9: bears on the output horn face (the plate itself clears the parent servo's shell by 0.5).
3. Output plate holes, through: ⌀7 centre (horn centre screw) + 4X ⌀3.4 on ⌀14 PCD at 45°.
4. Idler plate (-Y), y -22.9 to -18.9: circle R10.5 + neck polygon N1 to N4 (mirrored). The neck clears the parent servo's bus plugs.
5. Idler plate pad ⌀19 on the inner face, y -18.9 to -18.55, bearing on the passive horn; centre ⌀7 through (clears the idler boss); 4X ⌀3.4 on ⌀14 PCD at 45°, counterbore ⌀6.5 0.8 deep from outside (4X M3 × 5 into the passive horn, which turns on the boss); ⌀8.6 0.8 deep on the centre, and the two counterbores at x < 0 run out through the hub rim (see TAL-110).
6. Clevis gap in the link body: remove a cylinder R21 about Y between y ±18.9, and everything at x < 0 between those planes, except the two horn pads (dimensioned on each link's section).

NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.



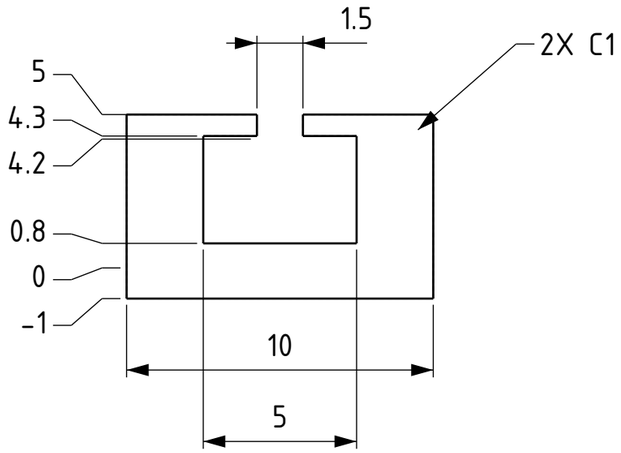
FRONT (IDLER PLATE, -Y)



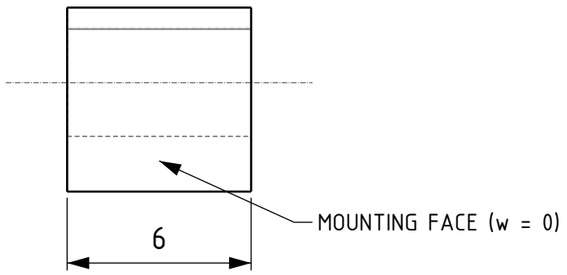
REAR (OUTPUT PLATE, +Y)

THIRD ANGLE PROJECTION

REFERENCE SHEET: in work sets A and B	
PROJECT TALOS 5-axis arm	EST. MASS -
PART Common features: clevis (part sheet 1 of 2)	SCALE 2:1
PART NO. TAL-002	UNITS / REV mm / A
QTY / ARM -	DATE 2026-11-09
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 4 of 21



CABLE CLIP: END VIEW



CABLE CLIP: SIDE VIEW

SEAT S (FROM THE SERVO MID-PLANE)

PART	FACE	S
arm shells, gripper	top	18.1
arm shells, gripper	bottom	17.9
shoulder	both	18.5
base	bottom	18.4
wrist	bottom	18.0

COMMON FEATURES

CABLE CLIP, one bus cable. Local u along the cable, v across, w out of the mounting face.

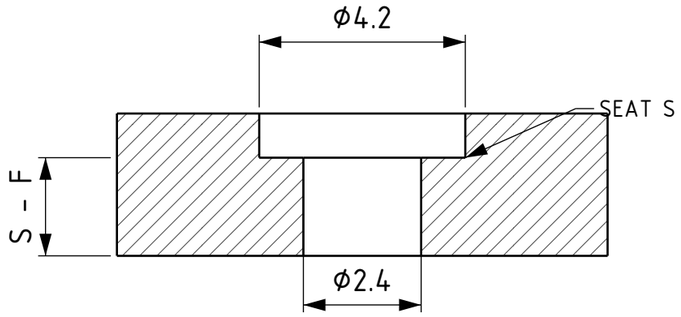
1. Block u ±3, v ±5, w −1 to 5 (sunk 1 into the face).
2. Tunnel through along u: v ±2.5, w 0.8 to 4.3. Slit v ±0.75 from w 4.2 out through the top.
3. Chamfer C1 on the two outer long edges.

CASE SCREW HOLE (M2 self-tap into the servo case): ⌀2.4 from the servo face F to the seat S, then ⌀4.2 counterbore from S outward through the part. Screw length: S − F + 3.5, rounded down.

HEX WINDOW H(xa, xb, zc, k): hexagon (xa, zc) (xa+k, zc−k) (xb−k, zc−k) (xb, zc) (xb−k, zc+k) (xa+k, zc+k), cut through the part along Y. Drawn for H(0, 40, 0, 10).

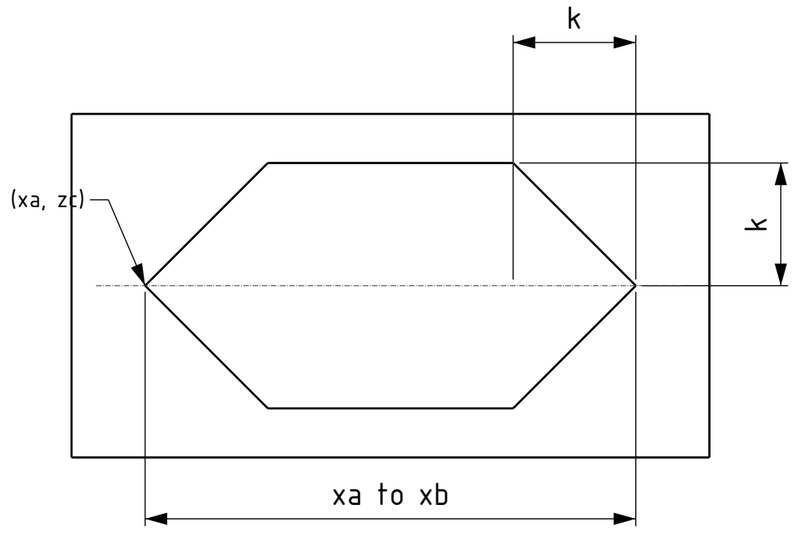
NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.



CASE SCREW HOLE (SECTION) SCALE 8:1

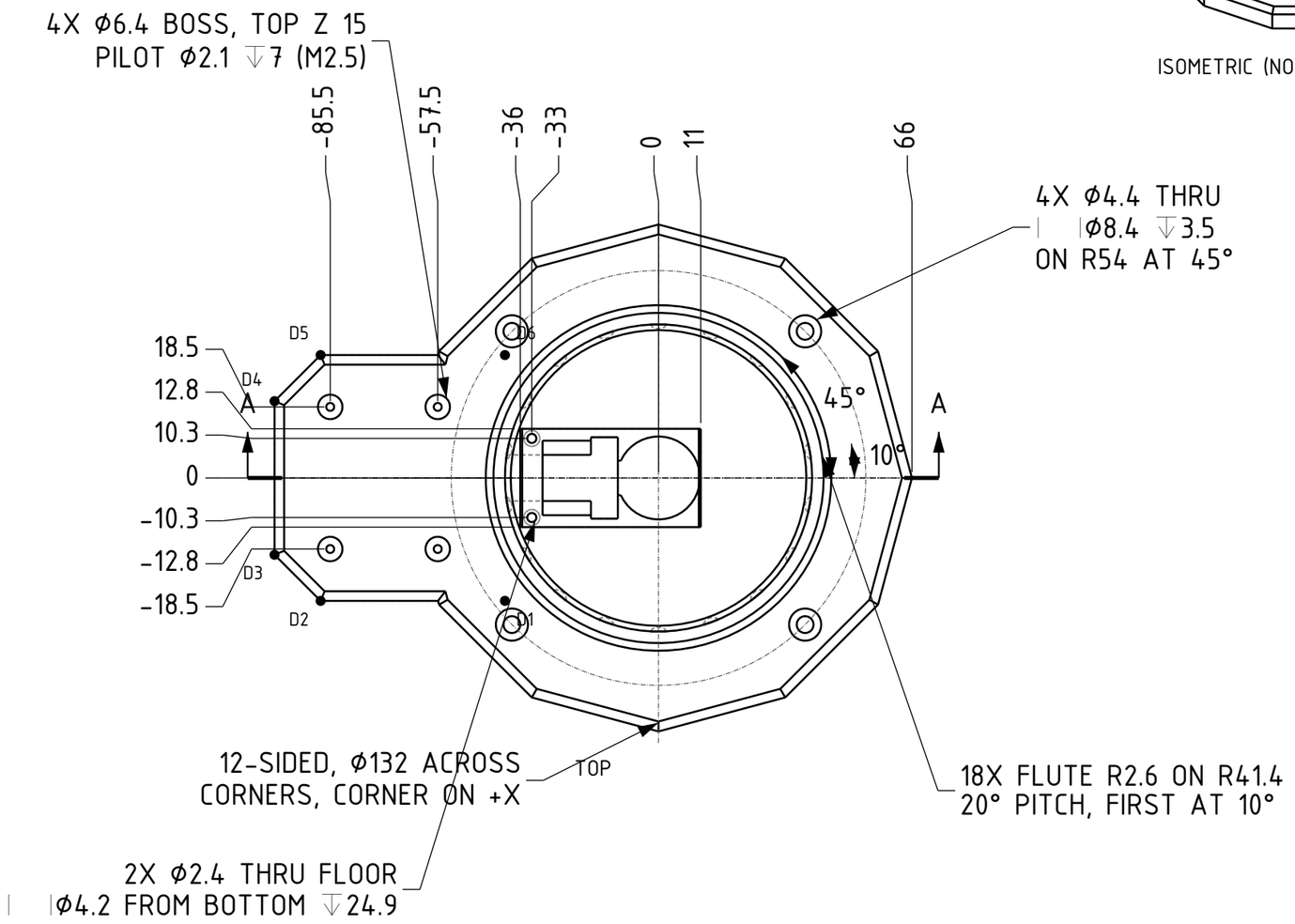
SERVO FACE F: TOP FACE 15.5,
BOTTOM FACE 16.5, FROM THE SERVO MID-PLANE

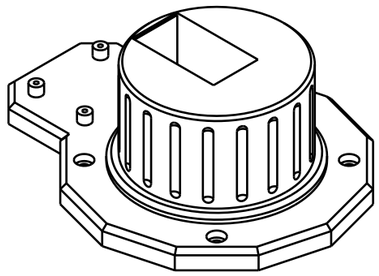


HEX WINDOW H(xa, xb, zc, k) SCALE 2:1

THIRD ANGLE PROJECTION

REFERENCE SHEET: in work sets A and B	
PROJECT TALOS 5-axis arm	EST. MASS -
PART Common features: clip, screw hole, window (part sheet 2 of 2)	SCALE 5:1
PART NO. TAL-002	UNITS / REV mm / A
QTY / ARM -	DATE 2026-11-09
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 5 of 21





ISOMETRIC (NOT TO SCALE)

BOARD DECK POINTS (X, Y)

PT	X	Y
D1	-40	-32
D2	-88	-32
D3	-100	-20
D4	-100	20
D5	-88	32
D6	-40	32

FEATURES (in order)

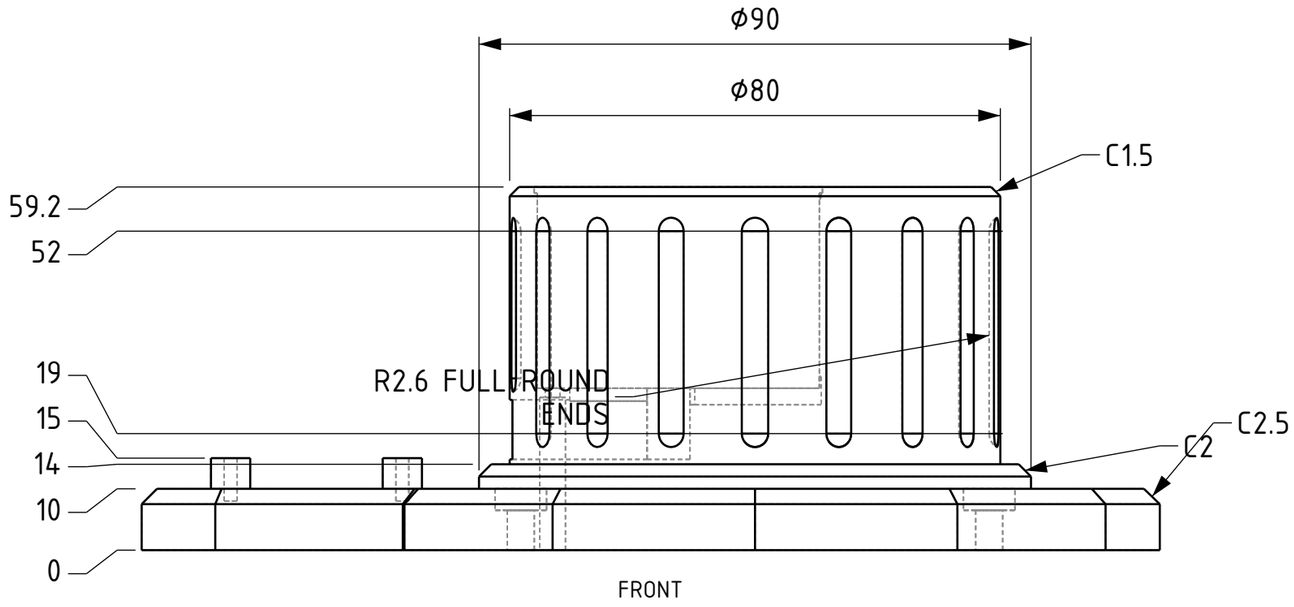
1. Plinth: regular 12-gon, ⌀132 across corners (a corner on +X), z 0 to 10.
2. Board deck: polygon D1 to D6, z 0 to 10; union with the plinth.
3. Chamfer C2.5 on all top outer edges (z 10).
4. Mounting holes: 4X ⌀4.4 through on R54 at 45° + 90°n; counterbore ⌀8.4 from z 6.5 up.
5. Collar ⌀90, z 10 to 14, C2 on its top edge.
6. Column ⌀80, z 10 to 59.2, C1.5 on its top edge.
7. Board standoffs: 4X ⌀6.4, z 9 to 15, at x -57.5 / -85.5, y ±18.5; pilot ⌀2.1 from z 8 to 16.
8. Flutes: 18X vertical ⌀5.2 on R41.4 at 10° + 20°n, axis z 19 to 52 (z 28 to 52 for the two at 170° and 190°, over the cable tunnel), spherical ends; cut.
9. Pocket P (sheet TAL-001) with its servo frame origin P0 at (0, 0, 43.3): servo +Z (shaft) along +Z, servo +X along +X, servo +Y along +Y.
10. Top opening: cut box x -36 to 11, y ±12.8, from z 58.2 up (the way in: the servo drops in from the top, connector end first).
11. Cable tunnel (plug bay to board deck): cut box x -60 to -13, y ±6, z 14.8 to 24.5. Pocket P's bump channel (z down to 24.3) opens into its roof.
12. Case screw holes from below (TAL-002, seat 18.4) at x -33, y ±10.3: ⌀2.4 from z 24.9 up to the pocket floor (z 26.4), ⌀4.2 counterbore from the bottom to z 24.9.

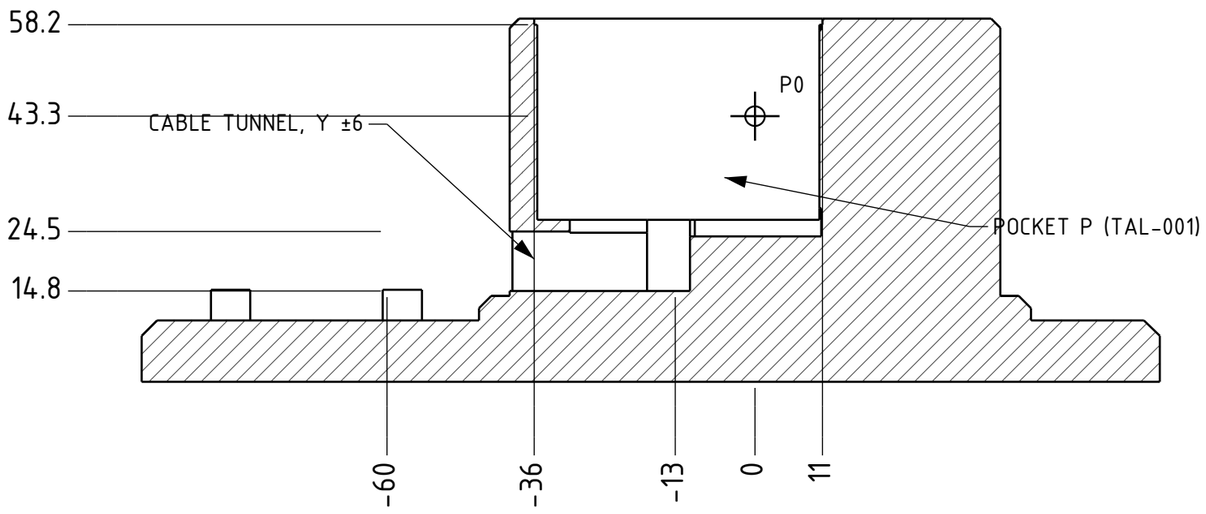
NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.
- The pan servo drops in from the top (its horns may stay on); its plugs hang into the bay under it. 2X M2 × 5 case screws from below.
- Print: plinth down; no support.

THIRD ANGLE PROJECTION

WORK SET A: base, shoulder, upper arm, hub caps, fit gauge	
PROJECT TALOS 5-axis arm	EST. MASS 154 g
PART Base (part sheet 1 of 2)	SCALE 2:3
PART NO. TAL-101	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-09
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 6 of 21





SECTION A-A

SECTION A-A

Section on the XZ plane (cutting line on sheet 1), looking toward +Y.

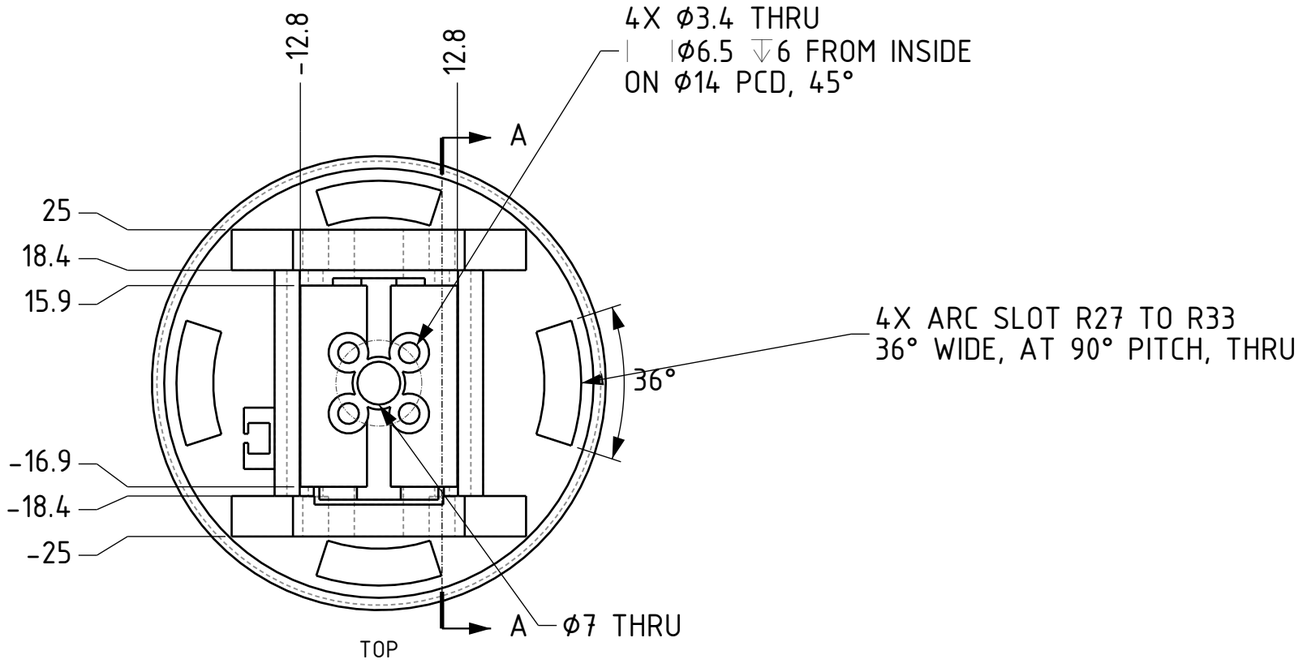
P0 = pocket P servo frame origin; servo axes along the part axes.

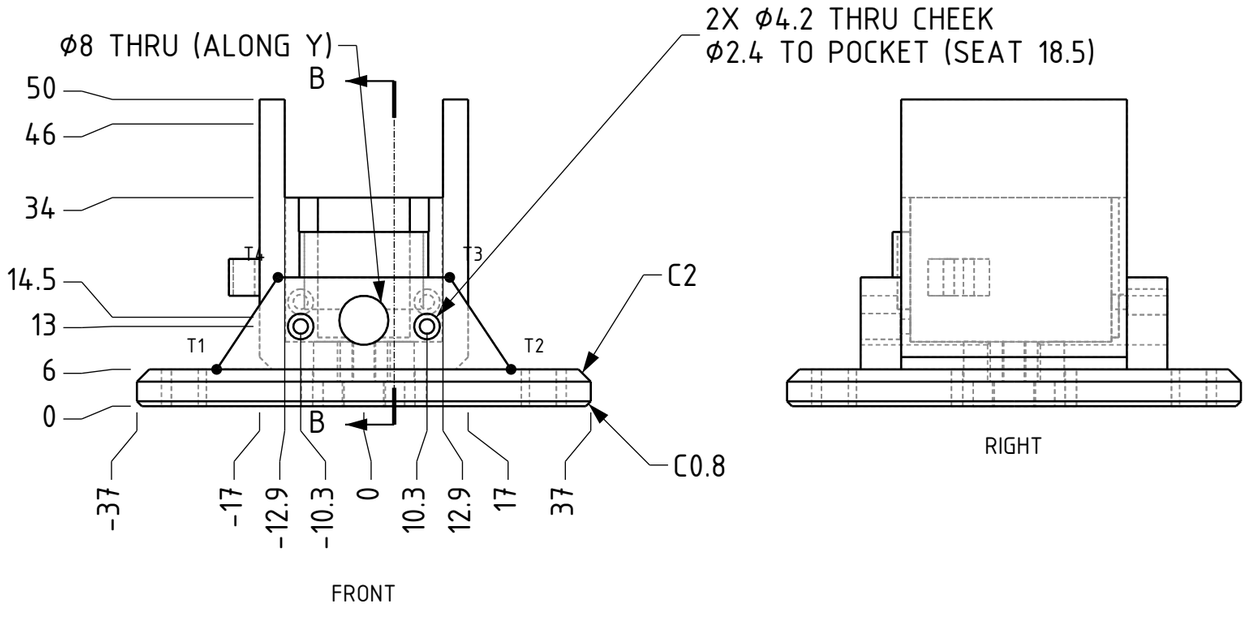
NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.

THIRD ANGLE PROJECTION

WORK SET A: base, shoulder, upper arm, hub caps, fit gauge	
PROJECT TALOS 5-axis arm	EST. MASS -
PART Base (part sheet 2 of 2)	SCALE 1:1
PART NO. TAL-101	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-09
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 7 of 21





CHEEK PROFILE (X, Z)

PT	X	Z
T1	-24	6
T2	24	6
T3	14	21
T4	-14	21

FEATURES (in order)

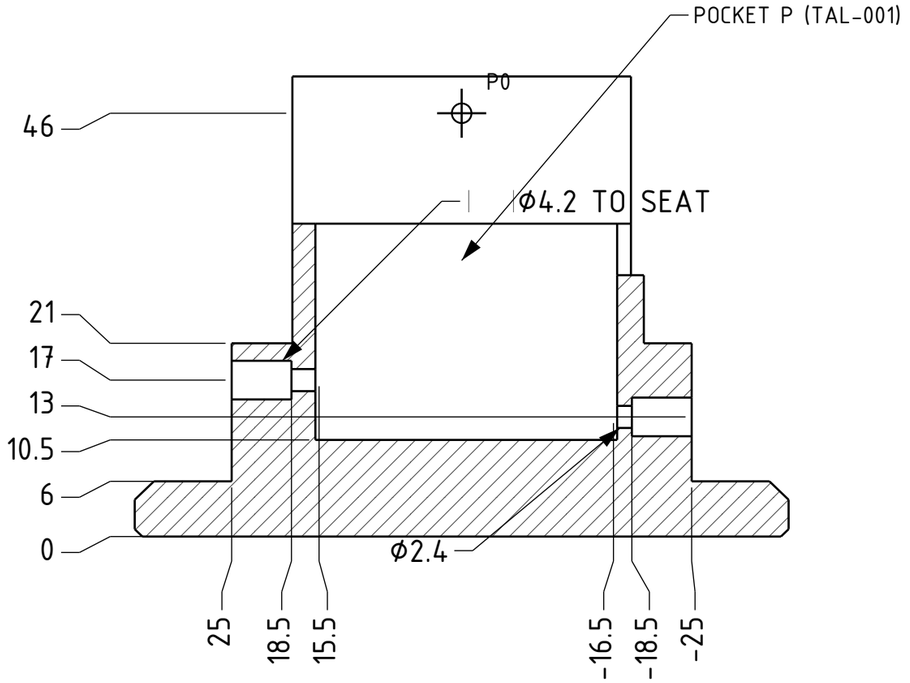
1. Turntable Ø74, z 0 to 6; C2 top edge, C0.8 bottom edge.
2. Cradle: box x ±17, y ±18.4, z 5 to 34.
3. Posts: boxes x −17 to −12.9 and 12.9 to 17, y ±18.4, z 30 to 50.
4. Cheeks: profile T1 to T4, y 18.4 to 25 and −25 to −18.4 (fused to the cradle walls).
5. Doubler on the −Y cradle wall: box x ±10.5, y −19.8 to −18.4, z 15 to 34 (leaves 0.8 over the bump channel).
6. Cable clip (TAL-002) on the −X face: centre (−17, −9, 21), cable along Z.
7. Pocket P (sheet TAL-001) with its servo frame origin P0 at (0, 0, 46): servo +Z (shaft) along +Y, servo +X along +Z (body hangs down), servo +Y along +X.
8. Servo opening: cut box x ±12.8, y −16.9 to 15.9, from z 10.5 up. Way in from the top (pocket P run on up, TAL-001; horns off): +Y wall: rib channel x ±7.5, y 15.5 to 17.1, from z 11.2 up; spline slot x ±3.5, y to 17.6, from z 46 up. −Y wall: bump channel x ±9.7, y −19 to −16.5, from z 15.8 up; plug slot through the wall x ±10.6, from z 28.4 up; boss slot x ±3.5, from z 46 up.
9. Cable hole Ø8 along Y, through all, at x 0, z 14.
10. Pan horn: Ø7 centre through. 4X Ø3.4 through on Ø14 PCD at 45°, counterbore Ø6.5 from the floor (z 10) down to z 4 (M3 × 6 from inside the cradle); centre opened to Ø8.6 over the same depth (no thin web between the counterbores and the centre hole).
11. Turntable windows: 4X arc slot R27 to R33, ±18° about 0°, 90°, 180°, 270°, through.
12. Case screw holes (TAL-002, seat 18.5) through the cheeks: +Y cheek at x ±10.3, z 17; −Y cheek at x ±10.3, z 13.

NOTES

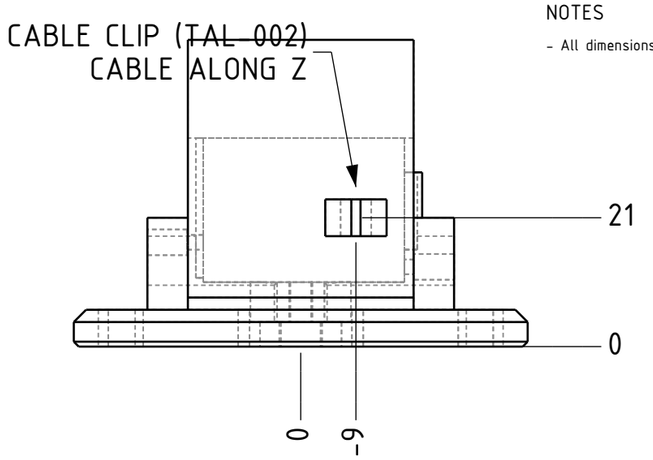
- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/−0.2 unless noted; servo pocket clearances are already included.
- Fit the shoulder servo from the top, far end first with the shaft to +Y, with both horns off; then the case screws through the cheeks (2X M2 × 6 on +Y, 2X M2 × 5 on −Y).
- Print: turntable down; no support.

THIRD ANGLE PROJECTION

WORK SET A: base, shoulder, upper arm, hub caps, fit gauge	
PROJECT TALOS 5-axis arm	EST. MASS 37 g
PART Shoulder (turntable + cradle) (part sheet 1 of 2)	SCALE 1:1
PART NO. TAL-102	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-09
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 8 of 21



SECTION A-A



LEFT SCALE 1:1

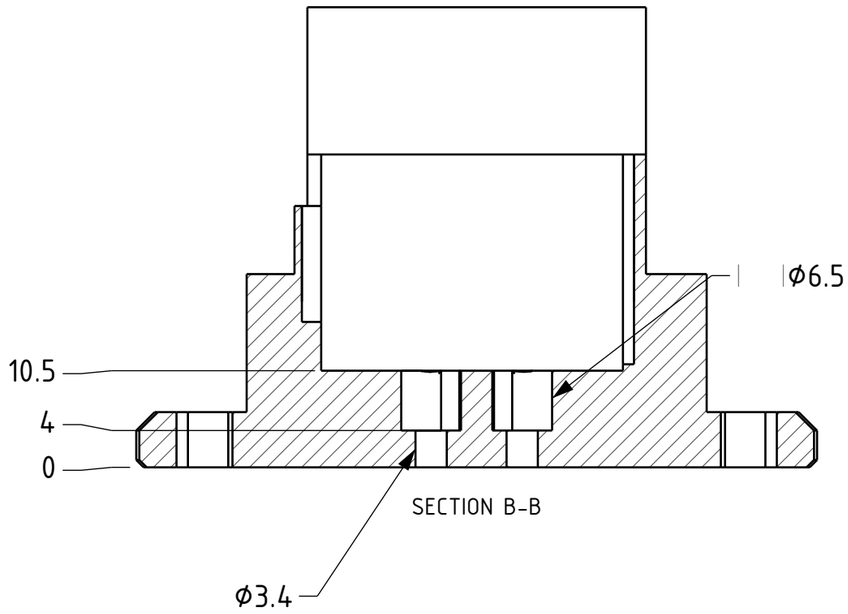
SECTIONS

A-A: plane x 10.3 through one hole of each case screw row (cutting line on sheet 1, top view), looking toward +X. The other holes mirror at x -10.3.

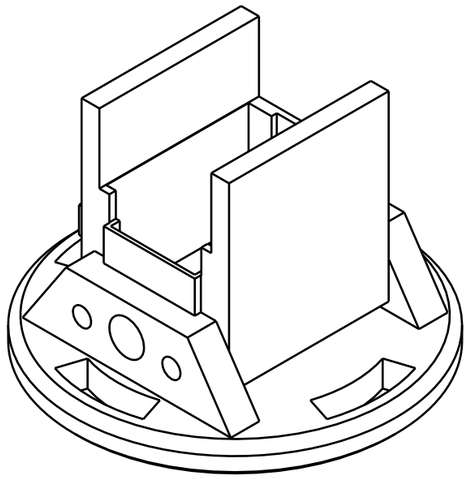
B-B: plane x 4.95 through two pan horn screw holes (cutting line on sheet 1, front view).

NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.



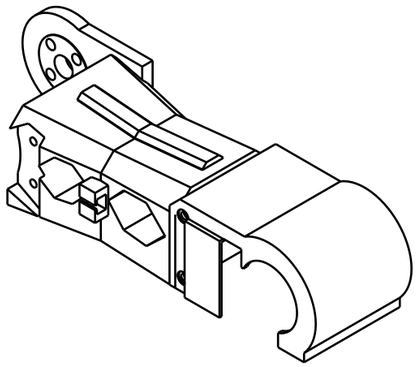
SECTION B-B



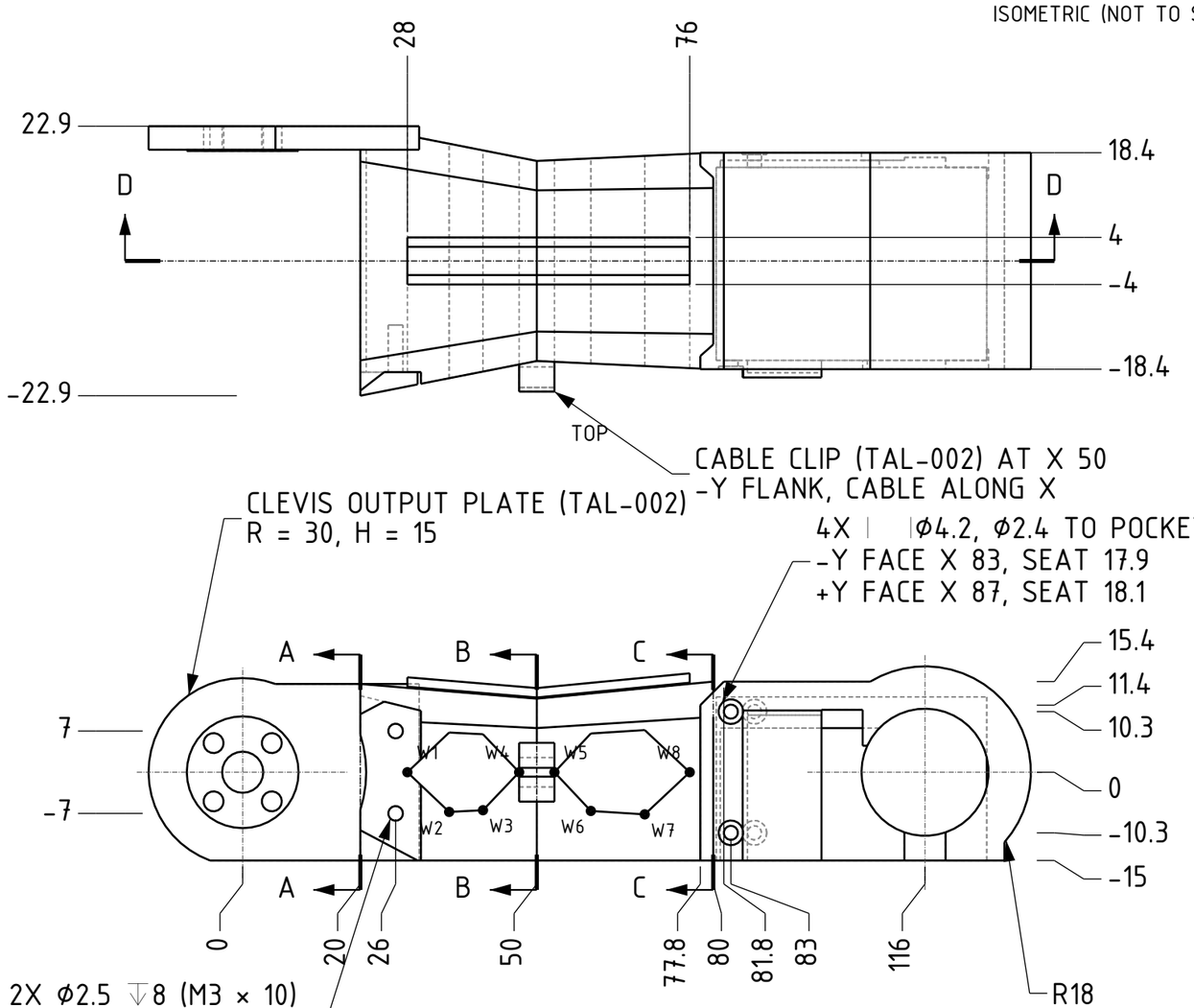
ISOMETRIC (NOT TO SCALE)

THIRD ANGLE PROJECTION

WORK SET A: base, shoulder, upper arm, hub caps, fit gauge	
PROJECT TALOS 5-axis arm	EST. MASS -
PART Shoulder (turntable + cradle) (part sheet 2 of 2)	SCALE 3:2
PART NO. TAL-102	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-09
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 9 of 21



ISOMETRIC (NOT TO SCALE)



WINDOW POINTS (X, Z), THROUGH Y

PT	X	Z
W1	28	0
W2	35.1	±6.75
W3	40.85	±6.46
W4	47	0
W5	53	0
W6	59.19	±6.58
W7	68.35	±7.16
W8	76	0

FEATURES (in order)

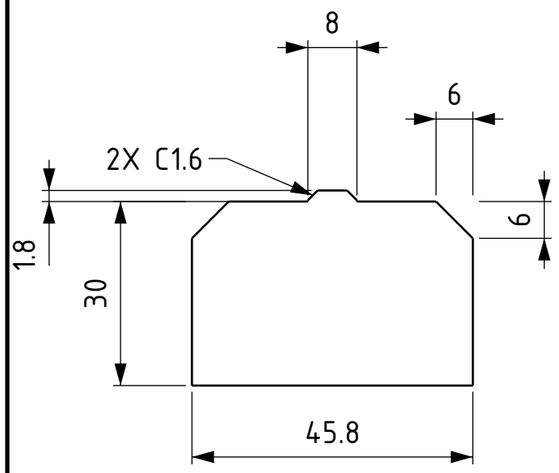
1. Clevis output plate only (TAL-002, +Y plate with its horn pad), R = 30, H = 15. The idler plate is the separate part TAL-110.
2. Beam: ruled loft of the profiles A, B, C (sheet 2) at x 20, 50, 80. Every profile sits on z −15, so the underside is flat; the waist dips on top only.
3. Servo shell: XZ polygon (77.8, −15) (116, −15) (116, 15.4) (81.8, 15.4) (77.8, 11.4), extruded y ±18.4; plus a cylinder R18 about Y at x 116.
4. Flat underside: cut everything below z −15 (trims the clevis hub and the R18 cylinder).
5. Nose floor: box x 125 to 129.5, y ±18.4, z −15 to −11; ties the shell walls together on the bed (the servo entry cut later trims its back to x 126.5).
6. Doubler on the −Y wall: box x 85 to 99.6, y −19.8 to −18.4, z −15 to 10.5 (leaves 0.8 over the bump channel; out of the next link's reach).
7. Spine: ruled loft of the ridge profiles A, B, C, trimmed to x 28 to 76.
8. Windows: polygons W1 to W4 and W5 to W8 (mirrored in Z), cut through Y.
9. Cable clip (TAL-002) on the −Y flank at x 50, cable along X.
10. Clevis gap cut (TAL-002): R21 about Y between y ±18.9, plus x < 0 between those planes; the output pad stays.
11. Pocket P (sheet TAL-001) with its servo frame origin P0 at (116, 0, 0): servo +Z (shaft) along +Y, servo +X along +X, servo +Y along −Z.
12. Servo entry from below (pocket P run on down, TAL-001; horns off): case box x 80.5 to 126.5, y −16.9 to 15.9, z below 12.8 (opens the floor under the whole servo). +Y wall: rib channel x 81.2 to 112.5, y 15.5 to 17.1, z below 7.5; spline slot x 116 ±3.5, y to 17.6, z below 0. −Y wall: bump channel x 85.8 to 98.8, y −19 to −16.5, z below 9.7; plug window x 98.4 to 105.4, z below 10.6; notch through the wall x 98.4 to 119.5, z below 0 (joins the plug window and the idler hole).
13. Case screw holes (TAL-002): +Y face (seat 18.1, over the rib channel; heads stand 0.7 proud) at x 87, z ±10.3; −Y face (seat 17.9) at x 83, z ±10.3.
14. Idler plate pocket: the TAL-110 outline (circle R10.5 + neck N1 to N4) grown 0.3, cut from y −23.9 to −18.9.
15. Idler plate screw holes: 2X ⌀2.5 at x 26, z ±7, from y −18.9 8 deep (M3 × 10).

NOTES

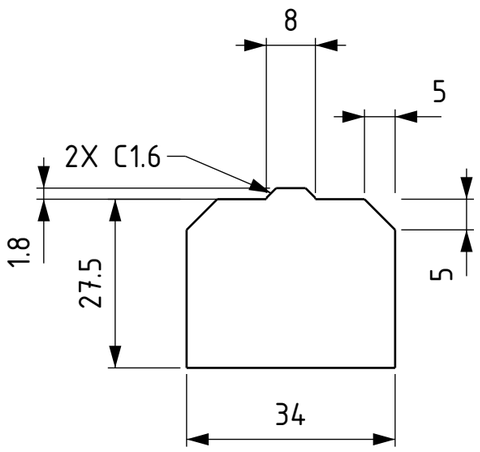
- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/−0.2 unless noted; servo pocket clearances are already included.
- Fit the elbow servo from below, shaft to +Y, with both horns off; then the case screws (2X M2 × 6 on +Y, 2X M2 × 4 on −Y) and refit the horns.
- The elbow servo's bus plugs come out through the −Y shell wall.
- Print: flat underside down; no support (servo-opening roof and window tops bridge). The flat underside is z −15.

THIRD ANGLE PROJECTION

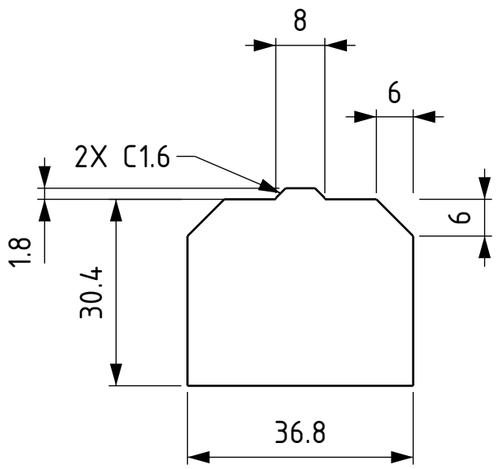
WORK SET A: base, shoulder, upper arm, hub caps, fit gauge	
PROJECT TALOS 5-axis arm	EST. MASS 48 g
PART Upper arm (part sheet 1 of 2)	SCALE 1:1
PART NO. TAL-103	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-09
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 10 of 21



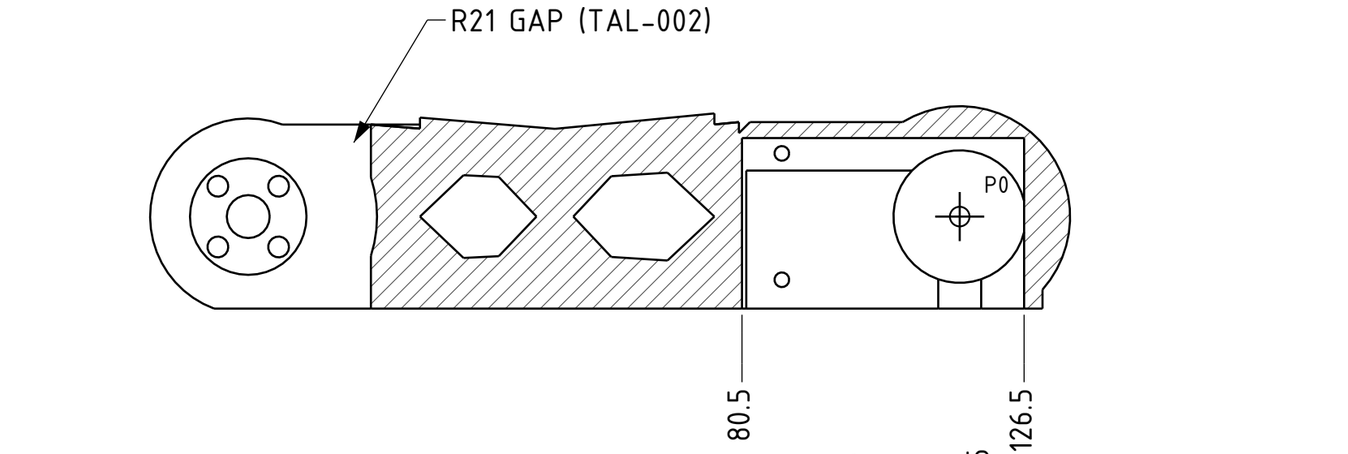
PROFILE A–A (X 20)



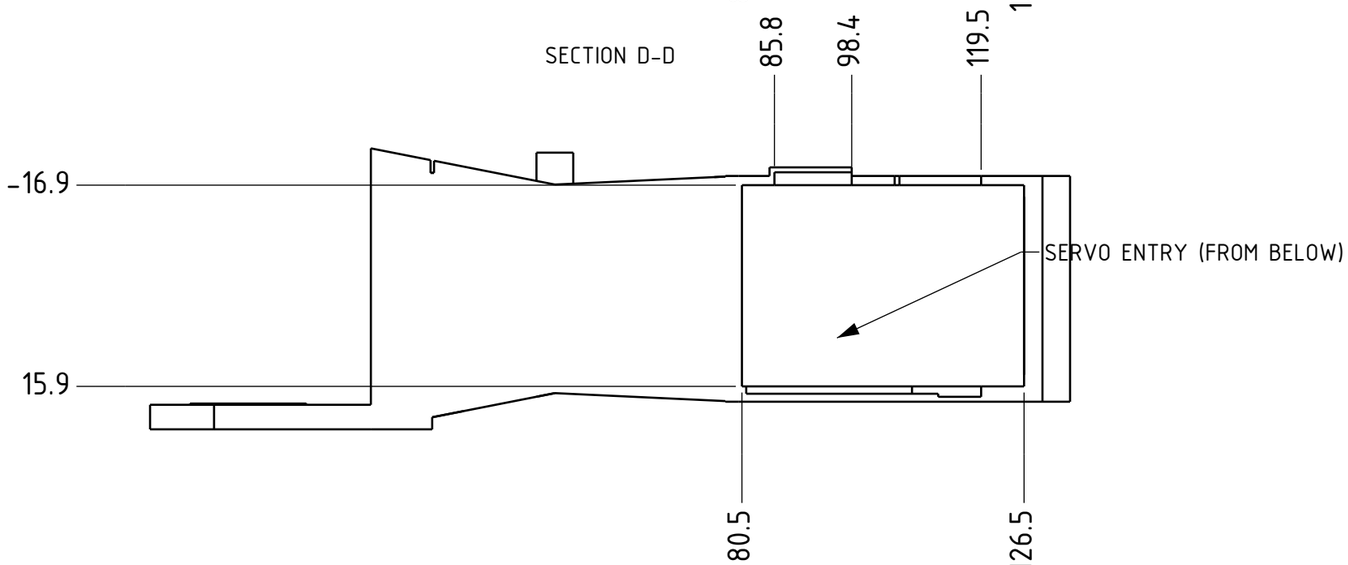
PROFILE B–B (X 50)



PROFILE C–C (X 80)



SECTION D–D



BOTTOM

PROFILES AND SECTION

Profiles A, B, C: beam section (top corners chamfered, underside flat and square-cornered) with the spine ridge on top, in the YZ plane at the loft stations; cutting lines on sheet 1. Each sits on z −15 (flat underside) with its top at the height given.

Ridge: 8 wide, 4.8 across the top, 1.8 above the beam, 45° bevels; its base runs 3 into the beam.

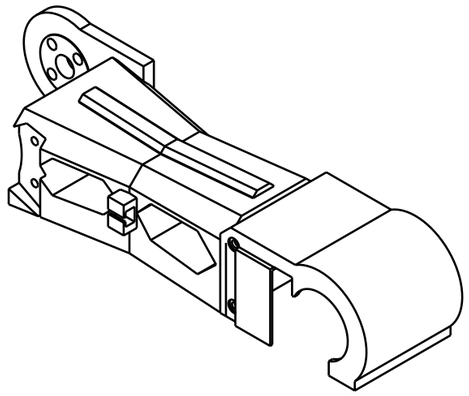
Section D–D: XZ plane, looking toward +Y. P0 = pocket P servo frame origin.

NOTES

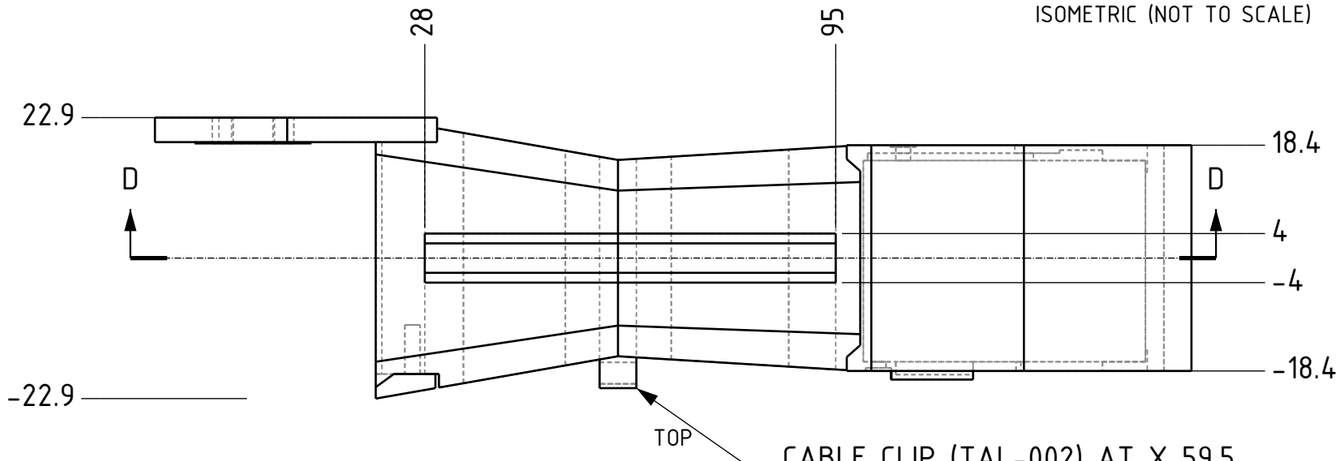
- All dimensions mm. Third-angle projection. Do not scale the drawing.

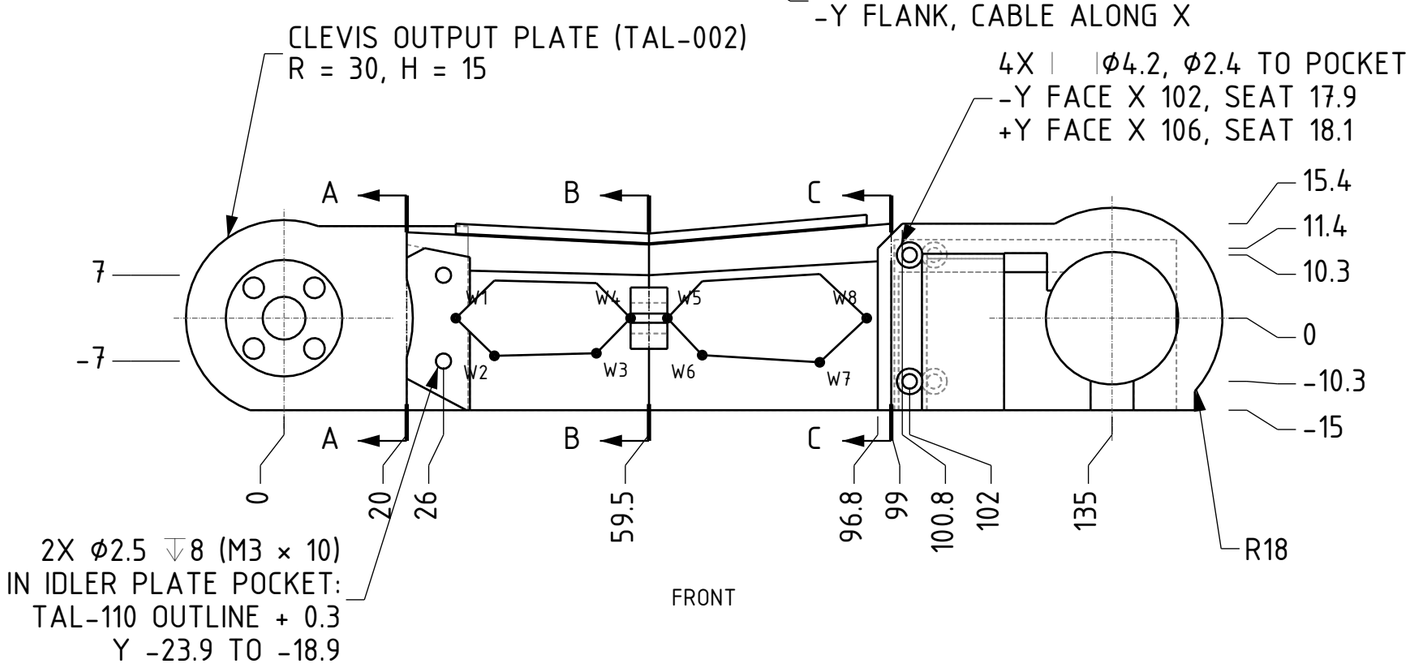
THIRD ANGLE PROJECTION

WORK SET A: base, shoulder, upper arm, hub caps, fit gauge	
PROJECT TALOS 5-axis arm	EST. MASS –
PART Upper arm (part sheet 2 of 2)	SCALE 1:1
PART NO. TAL-103	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-09
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 11 of 21



ISOMETRIC (NOT TO SCALE)





FRONT

WINDOW POINTS (X, Z), THROUGH Y

PT	X	Z
W1	28	0
W2	34.3	±6.14
W3	50.92	±5.72
W4	56.5	0
W5	62.5	0
W6	68.18	±6.03
W7	87.34	±7.19
W8	95	0

FEATURES (in order)

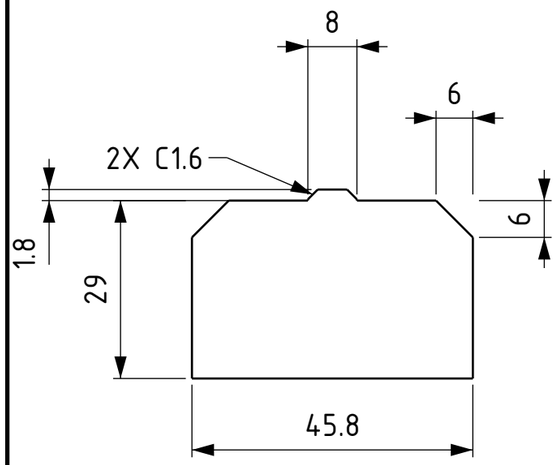
1. Clevis output plate only (TAL-002, +Y plate with its horn pad), R = 30, H = 15. The idler plate is the separate part TAL-110.
2. Beam: ruled loft of the profiles A, B, C (sheet 2) at x 20, 59.5, 99. Every profile sits on z -15, so the underside is flat; the waist dips on top only.
3. Servo shell: XZ polygon (96.8, -15) (135, -15) (135, 15.4) (100.8, 15.4) (96.8, 11.4), extruded y ±18.4; plus a cylinder R18 about Y at x 135.
4. Flat underside: cut everything below z -15 (trims the clevis hub and the R18 cylinder).
5. Nose floor: box x 144 to 148.5, y ±18.4, z -15 to -11; ties the shell walls together on the bed (the servo entry cut later trims its back to x 145.5).
6. Doubler on the -Y wall: box x 104 to 118.6, y -19.8 to -18.4, z -15 to 10.5 (leaves 0.8 over the bump channel; out of the next link's reach).
7. Spine: ruled loft of the ridge profiles A, B, C, trimmed to x 28 to 95.
8. Windows: polygons W1 to W4 and W5 to W8 (mirrored in Z), cut through Y.
9. Cable clip (TAL-002) on the -Y flank at x 59.5, cable along X.
10. Clevis gap cut (TAL-002): R21 about Y between y ±18.9, plus x < 0 between those planes; the output pad stays.
11. Pocket P (sheet TAL-001) with its servo frame origin P0 at (135, 0, 0): servo +Z (shaft) along +Y, servo +X along +X, servo +Y along -Z.
12. Servo entry from below (pocket P run on down, TAL-001; horns off): case box x 99.5 to 145.5, y -16.9 to 15.9, z below 12.8 (opens the floor under the whole servo). +Y wall: rib channel x 100.2 to 131.5, y 15.5 to 17.1, z below 7.5; spline slot x 135 ±3.5, y to 17.6, z below 0. -Y wall: bump channel x 104.8 to 117.8, y -19 to -16.5, z below 9.7; plug window x 117.4 to 124.4, z below 10.6; notch through the wall x 117.4 to 138.5, z below 0 (joins the plug window and the idler hole).
13. Case screw holes (TAL-002): +Y face (seat 18.1, over the rib channel; heads stand 0.7 proud) at x 106, z ±10.3; -Y face (seat 17.9) at x 102, z ±10.3.
14. Idler plate pocket: the TAL-110 outline (circle R10.5 + neck N1 to N4) grown 0.3, cut from y -23.9 to -18.9.
15. Idler plate screw holes: 2X ⌀2.5 at x 26, z ±7, from y -18.9 8 deep (M3 × 10).

NOTES

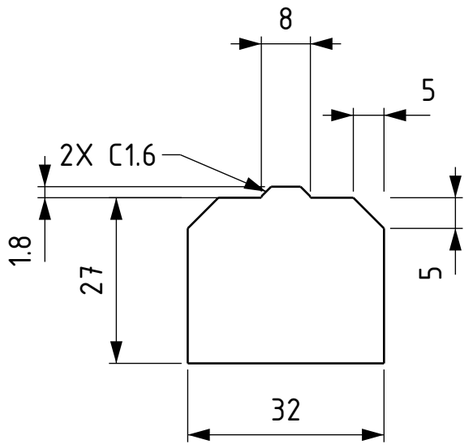
- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.
- Fit the wrist flex servo from below, shaft to +Y, with both horns off; then the case screws (2X M2 × 6 on +Y, 2X M2 × 4 on -Y) and refit the horns.
- The wrist flex servo's bus plugs come out through the -Y shell wall.
- Print: flat underside down; no support (servo-opening roof and window tops bridge). The flat underside is z -15.

THIRD ANGLE PROJECTION

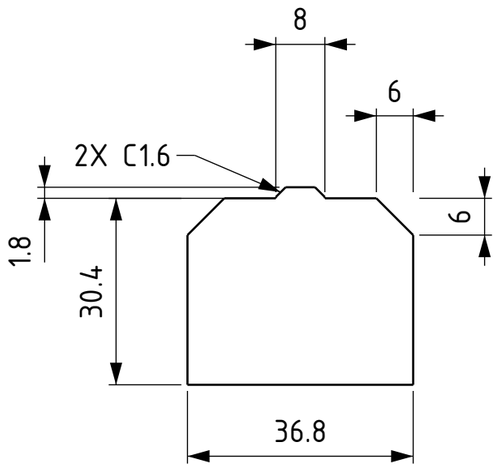
WORK SET B: forearm, wrist, gripper, moving jaw, idler plates	
PROJECT TALOS 5-axis arm	EST. MASS 53 g
PART Forearm (part sheet 1 of 2)	SCALE 1:1
PART NO. TAL-104	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-09
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 12 of 21



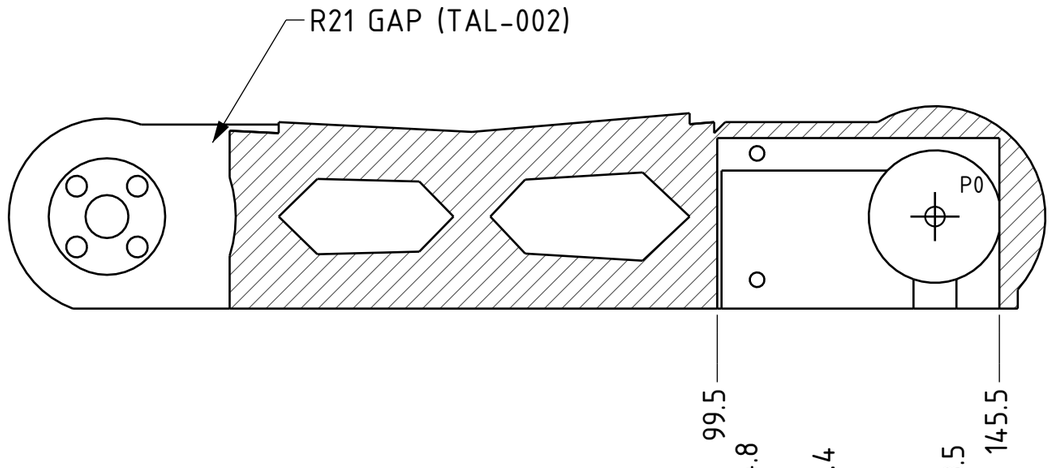
PROFILE A–A (X 20)



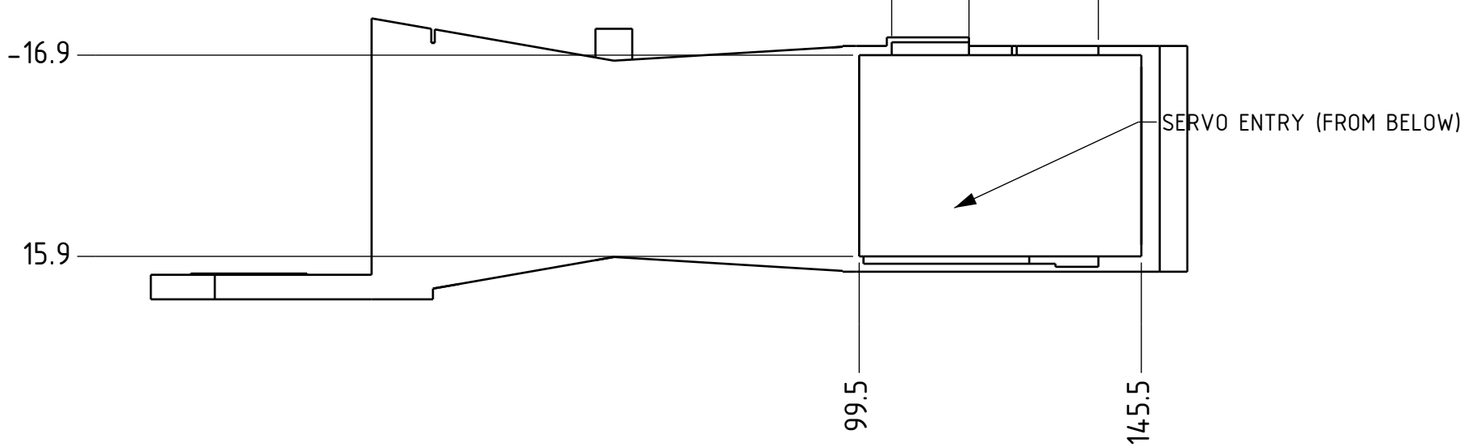
PROFILE B–B (X 59.5)



PROFILE C–C (X 99)



SECTION D–D



BOTTOM

PROFILES AND SECTION

Profiles A, B, C: beam section (top corners chamfered, underside flat and square-cornered) with the spine ridge on top, in the YZ plane at the loft stations; cutting lines on sheet 1. Each sits on z −15 (flat underside) with its top at the height given.

Ridge: 8 wide, 4.8 across the top, 1.8 above the beam, 45° bevels; its base runs 3 into the beam.

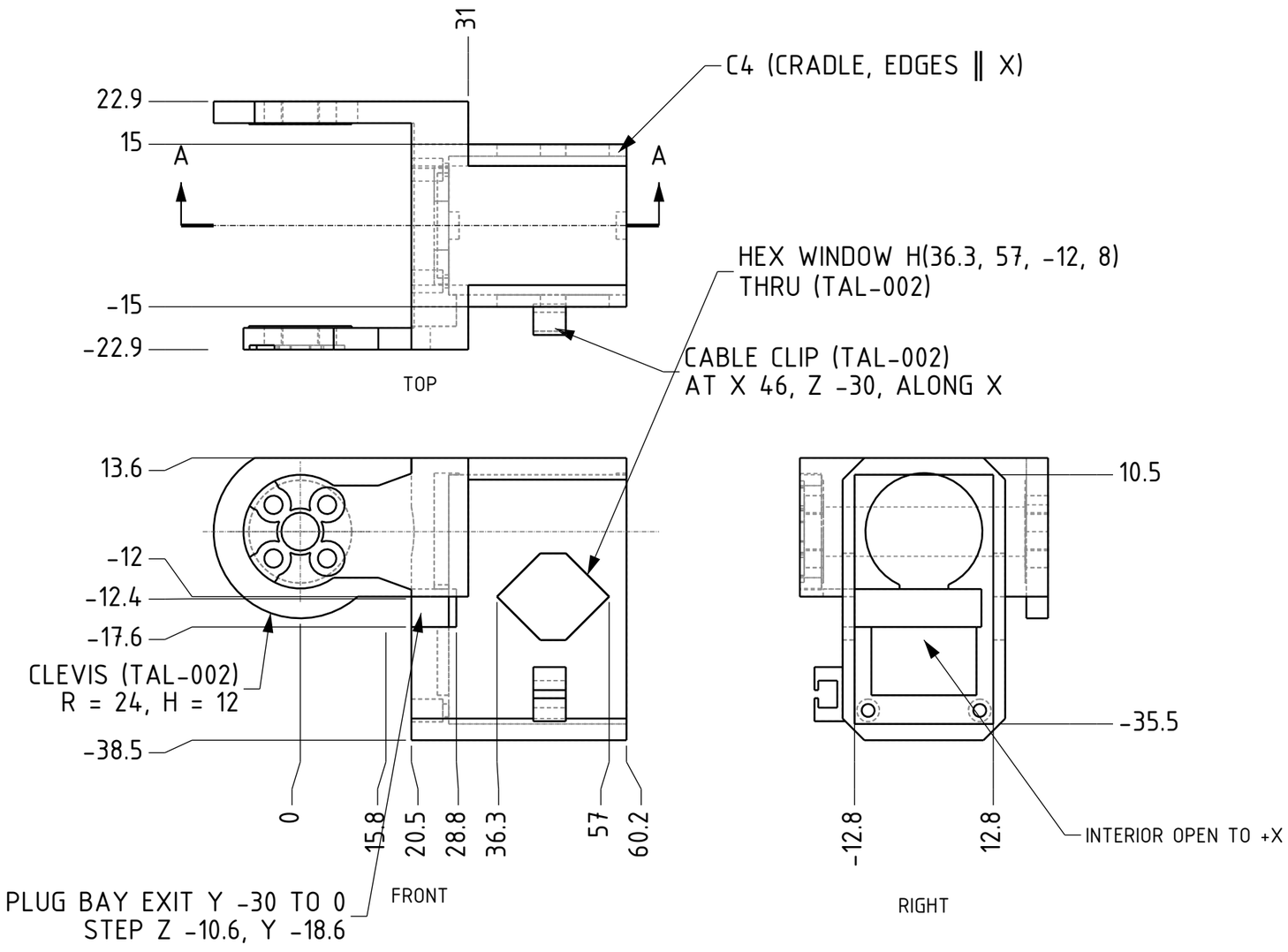
Section D–D: XZ plane, looking toward +Y. P0 = pocket P servo frame origin.

NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.

THIRD ANGLE PROJECTION

WORK SET B: forearm, wrist, gripper, moving jaw, idler plates	
PROJECT TALOS 5-axis arm	EST. MASS –
PART Forearm (part sheet 2 of 2)	SCALE 1:1
PART NO. TAL-104	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-09
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 13 of 21



FEATURES (in order)

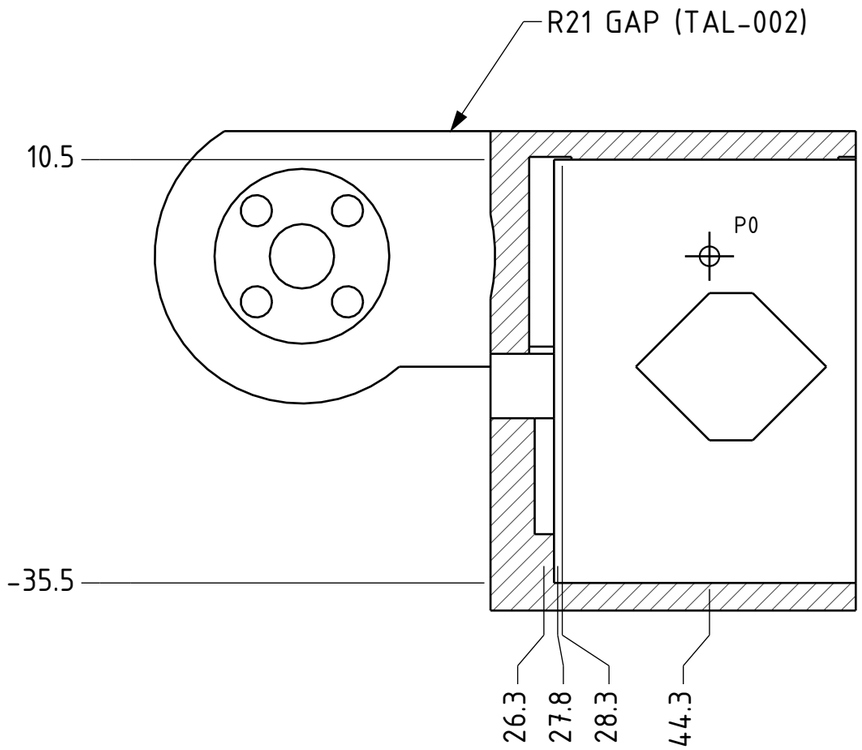
1. Clevis (TAL-002), R = 24, H = 12; the output plate's rectangle extended up to z 13.6, with the horn holes cut through it.
2. Bridge lugs: box x 20.5 to 31, y ±22.9, z -12 to 13.6.
3. Cradle: box x 20.5 to 60.2, y ±15, z -38.5 to 13.6; C4 on the edges parallel to X.
4. Flat top: cut everything above z 13.6 (trims the clevis hubs), so the wrist prints upside down on one face.
5. Cable clip (TAL-002) on the -Y face: centre (46, -15.2, -30), cable along X.
6. Clevis gap cut (TAL-002); both horn pads stay.
7. Pocket P (sheet TAL-001) with its servo frame origin P0 at (44.3, 0, 0): servo +Z (shaft) along +X, servo +X along +Z (body hangs down), servo +Y along -Y.
8. Cradle interior: cut box x 28.3 to the +X end, y ±12.8, z -35.5 to 10.5 (the way in: the servo slides in along the roll axis, connector end first; its horns may stay on).
9. Plug bay exit to -Y: cut boxes x 15.8 to 28.8: y -18.6 to 0, z -17.6 to -10.6; and y -30 to 0, z -17.6 to -12.4.
10. Side windows: hex window H(36.3, 57, -12, 8) through Y (TAL-002).
11. Case screw holes (TAL-002, seat 18.0), bottom face only, from the back wall: (y, z) = (±10.3, -33).

NOTES

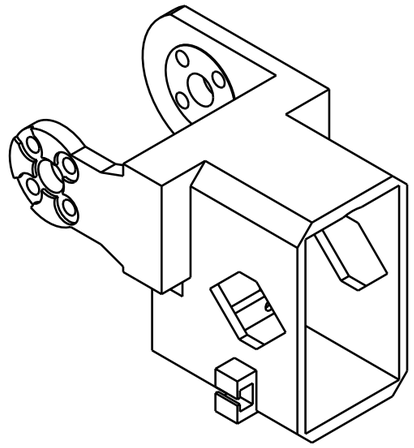
- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.
- Fit the roll servo's 2X M2 × 5 case screws before the gripper goes on.
- Print: upside down, flat top face down; support only under the idler plate's neck (5 mm up); the cradle floor and plug bay roof bridge.

THIRD ANGLE PROJECTION

WORK SET B: forearm, wrist, gripper, moving jaw, idler plates	
PROJECT TALOS 5-axis arm	EST. MASS 30 g
PART Wrist (part sheet 1 of 2)	SCALE 1:1
PART NO. TAL-105	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-09
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 14 of 21



SECTION A-A



ISOMETRIC (NOT TO SCALE)

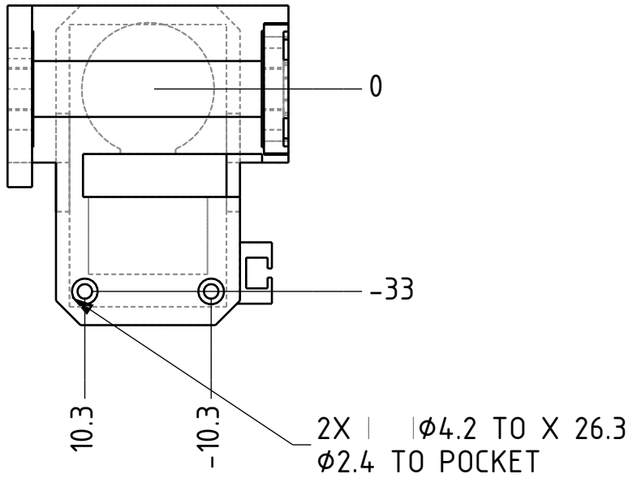
SECTION A-A AND LEFT VIEW

A-A: XZ plane (cutting line on sheet 1), looking toward +Y. P0 = pocket P servo frame origin.

Left view: the cradle back wall, seen from -X between the clevis plates.

NOTES

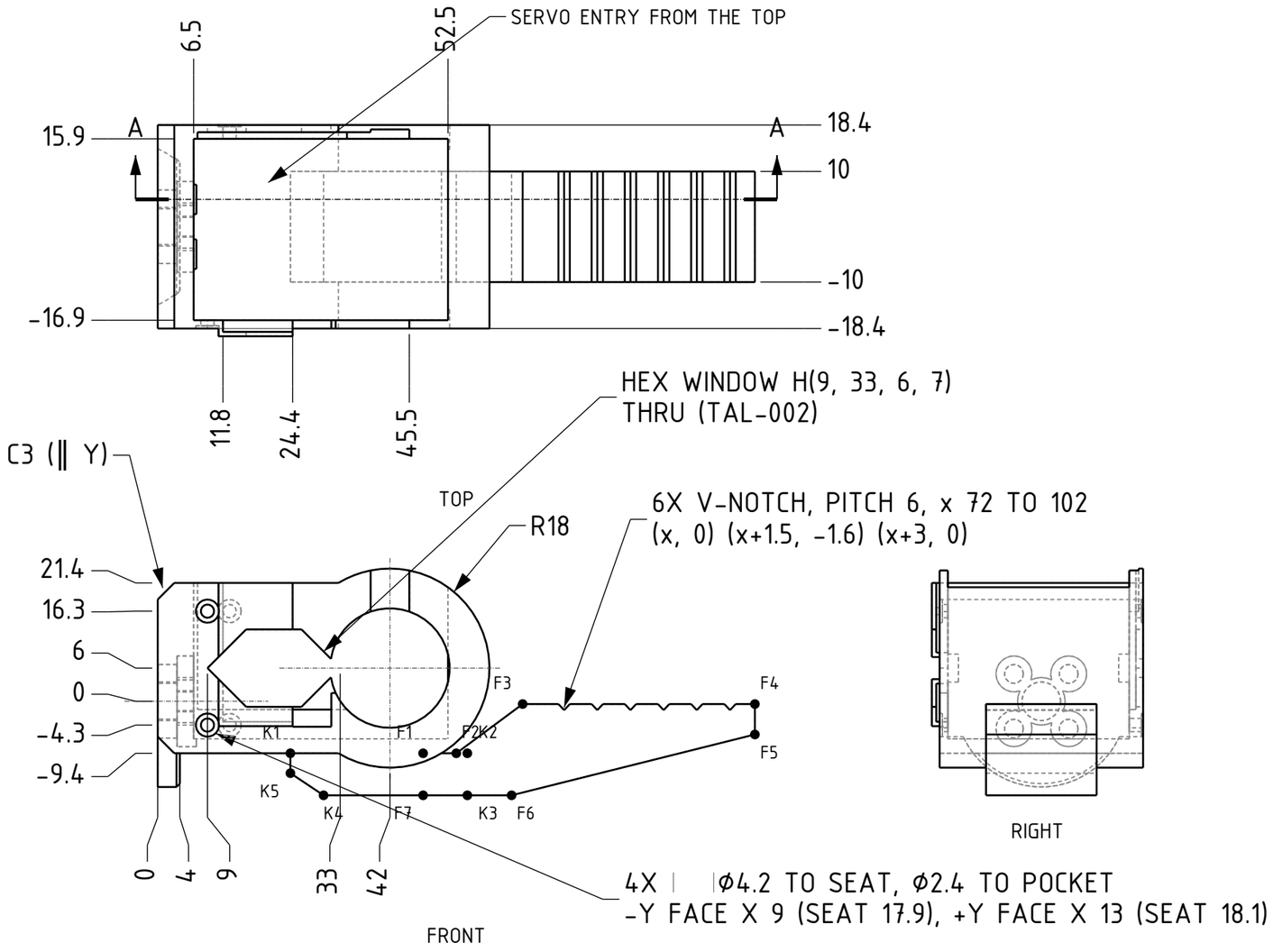
- All dimensions mm. Third-angle projection. Do not scale the drawing.



LEFT SCALE 1:1

THIRD ANGLE PROJECTION

WORK SET B: forearm, wrist, gripper, moving jaw, idler plates	
PROJECT TALOS 5-axis arm	EST. MASS -
PART Wrist (part sheet 2 of 2)	SCALE 3:2
PART NO. TAL-105	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-09
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 15 of 21



KEEL K AND FINGER F PROFILES (X, Z), Y ±10

PT	X	Z
K1	24	-9.4
K2	56	-9.4
K3	56	-17
K4	30	-17
K5	24	-13
F1	48	-9.4
F2	54	-9.4
F3	66	-0.5
F4	108	-0.5
F5	108	-6
F6	64	-17
F7	48	-17

FEATURES (in order)

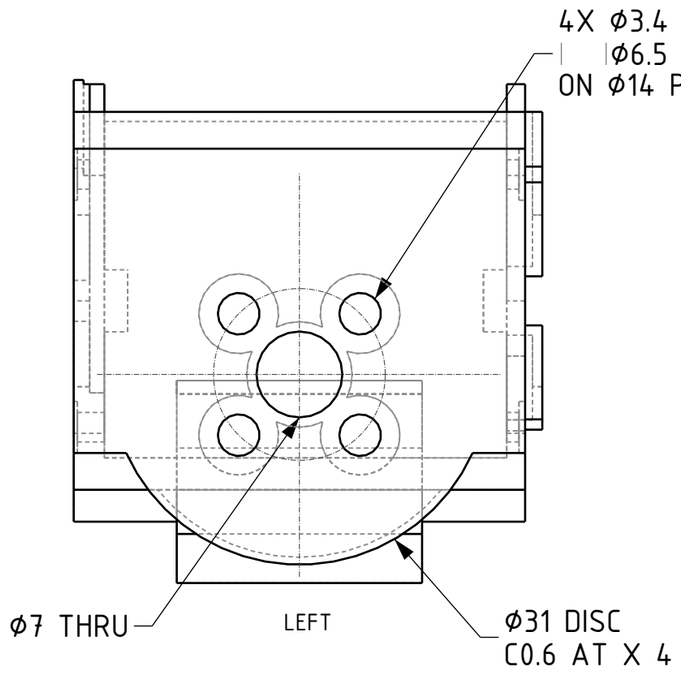
1. Disc Ø31 along X, x 0 to 4; C0.6 on the front edge.
2. Housing: box x 0 to 42, y ±18.4, z -9.4 to 21.4; C3 on the edges parallel to Y; plus a cylinder R18 about Y at (x, z) = (42, 6).
3. Keel K1 to K5 and finger F1 to F7, each extruded y ±10.
4. Doubler on the -Y wall: box x 11 to 25.6, y -19.8 to -18.4, z -4.5 to 21.4 (leaves 0.8 over the bump channel).
5. Pocket P (sheet TAL-001) with its servo frame origin P0 at (42, 0, 6): servo +Z (shaft) along +Y, servo +X along +X, servo +Y along -Z.
6. Servo entry from the top (pocket P run on up, TAL-001; horns off): case box x 6.5 to 52.5, y -16.9 to 15.9, z -6.8 up (opens the top over the whole servo). +Y wall: rib channel x 7.2 to 38.5, y 15.5 to 17.1, z -1.5 up; spline slot x 42 ±3.5, y to 17.6, z 6 up. -Y wall: bump channel x 11.8 to 24.8, y -19 to -16.5, z -3.7 up; plug window x 24.4 to 31.4, z -4.6 up; notch through the wall x 24.4 to 45.5, z 6 up (joins the plug window and the idler hole).
7. Side windows: hex window H(9, 33, 6, 7) through Y (TAL-002).
8. Grip notches: 6X triangle (x, 0) (x+1.5, -1.6) (x+3, 0), x = 72, 78 … 102, through Y.
9. Roll horn (sheet 2): Ø7 centre, x -1 to 7. 4X Ø3.4 on Ø14 PCD at 45°, x -1 to 3.5, counterbore Ø6.5 from x 3.5 to 7 (M3 × 6 from inside the housing); centre opened to Ø8.6 from x 3.5 to 7 (no thin web between the counterbores and the centre hole).
10. Case screw holes (TAL-002): +Y face (seat 18.1) at x 13, z -4.3 and 16.3; -Y face (seat 17.9) at x 9, z -4.3 and 16.3.

NOTES

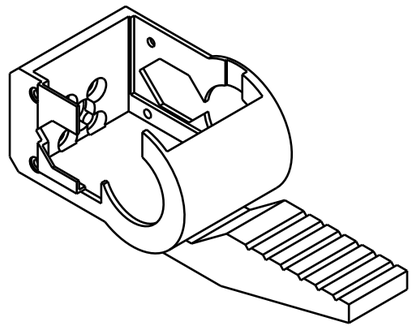
- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.
- Fit the jaw servo from the top, shaft to +Y, with both horns off; then the case screws (2X M2 × 6 on +Y, 2X M2 × 4 on -Y) and refit the horns.
- The jaw servo's bus plugs come out through the -Y housing wall.
- Print: standing on the roll-horn disc; no support (one 32 mm bridge at the front of the servo opening); use a brim, the part is 108 tall.

THIRD ANGLE PROJECTION

WORK SET B: forearm, wrist, gripper, moving jaw, idler plates	
PROJECT TALOS 5-axis arm	EST. MASS 29 g
PART Gripper body (with fixed jaw) (part sheet 1 of 2)	SCALE 1:1
PART NO. TAL-106	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-09
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 16 of 21



LEFT



ISOMETRIC (NOT TO SCALE)

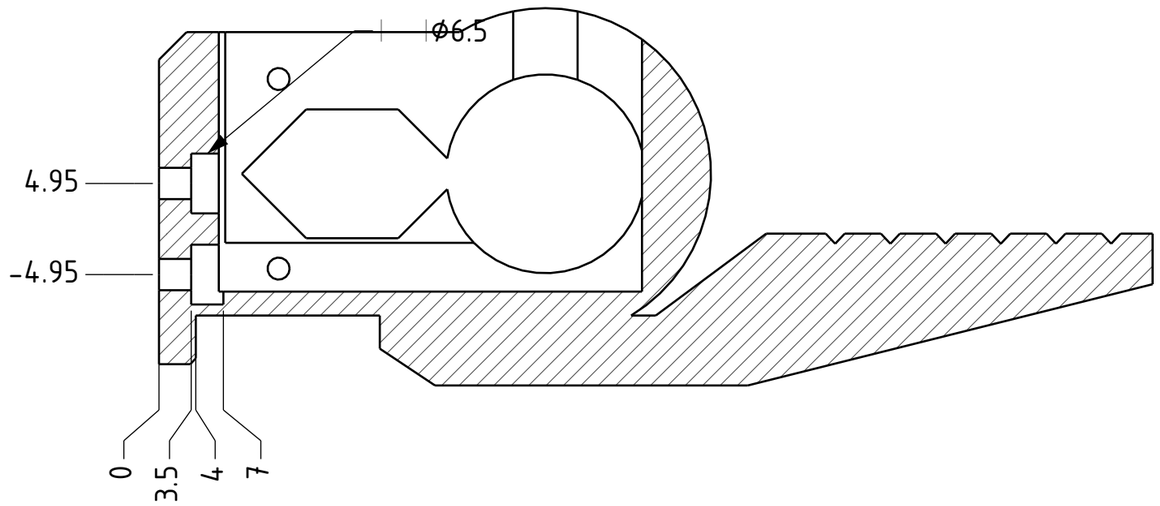
LEFT VIEW AND SECTION A-A

Left view: the disc face (roll horn side), seen from -X.

A-A: plane y 4.95 through two roll horn screws (cutting line on sheet 1, top view), looking toward +Y.

NOTES

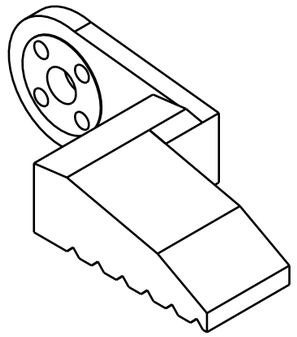
- All dimensions mm. Third-angle projection. Do not scale the drawing.



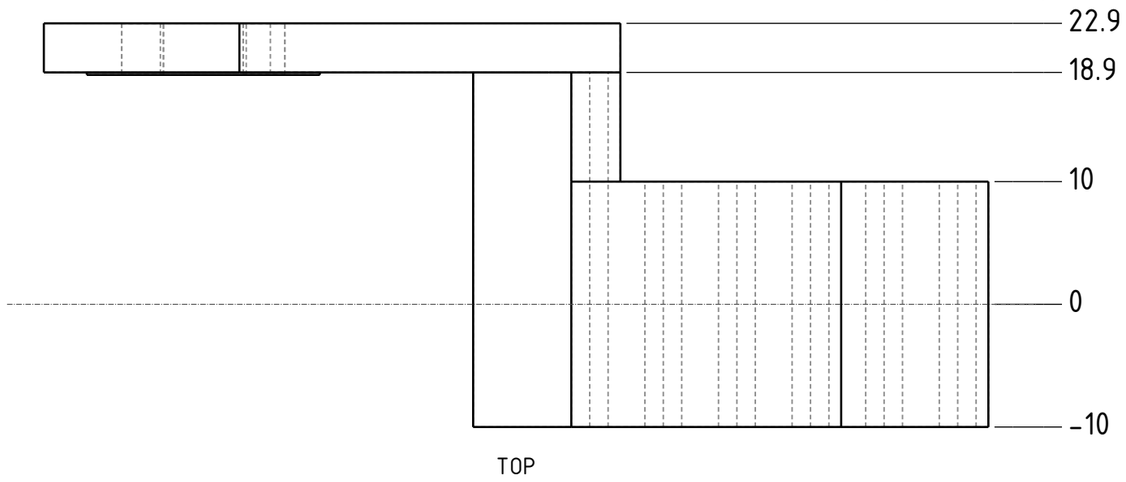
SECTION A-A SCALE 3:2

THIRD ANGLE PROJECTION

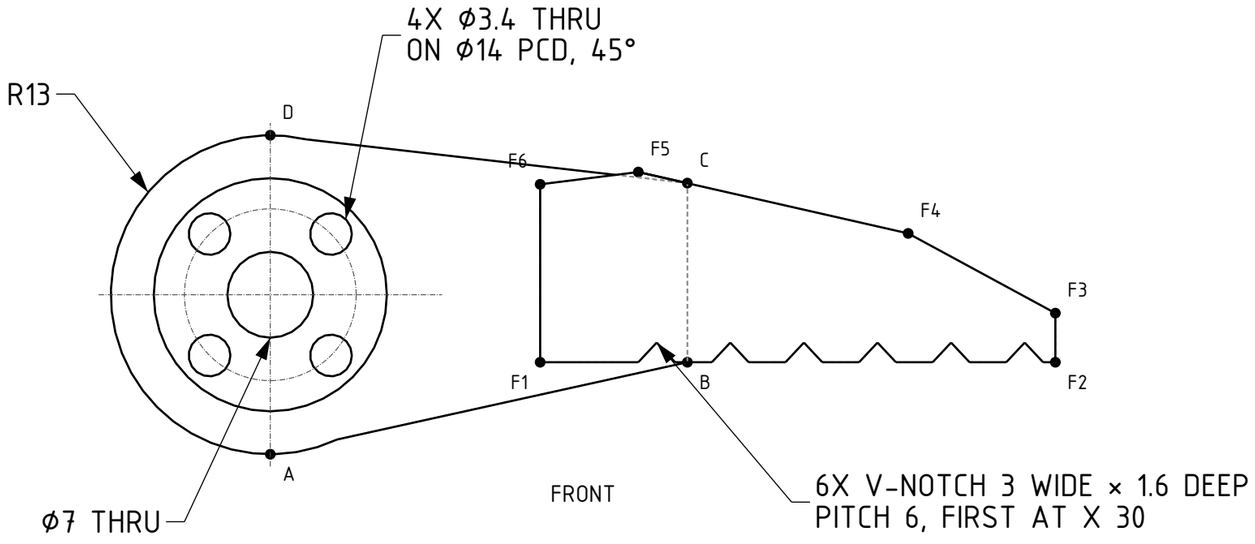
WORK SET B: forearm, wrist, gripper, moving jaw, idler plates	
PROJECT TALOS 5-axis arm	EST. MASS -
PART Gripper body (with fixed jaw) (part sheet 2 of 2)	SCALE 2:1
PART NO. TAL-106	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-09
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 17 of 21



ISOMETRIC (NOT TO SCALE)



TOP



FRONT

PROFILE POINTS (X, Z)

PT	X	Z	PROFILE
A	0	-13	plate
B	34	-5.5	plate, knuckle
C	34	9.1	plate, knuckle
D	0	13	plate
F1	22	-5.5	finger, knuckle
F2	64	-5.5	finger
F3	64	-1.5	finger
F4	52	5	finger
F5	30	10	finger, knuckle
F6	22	9	finger, knuckle

FEATURES (in order)

1. Plate, y 18.9 to 22.9: circle R13 at the origin + polygon A B C D.
2. Finger, y -10 to 10: polygon F1 F2 F3 F4 F5 F6.
3. Knuckle, y 10 to 18.9: polygon F1 B C F5 F6.
4. Grip notches: 6X V-notch on the finger's lower edge, y -10 to 18.9 (finger and knuckle; the plate is not notched), 3 wide × 1.6 deep, pitch 6, first at x 30.
5. Pad ⌀19 on the plate's inner face, y 18.7 to 18.9: bears on the output horn face.
6. Horn holes through the plate: ⌀7 centre + 4X ⌀3.4 on ⌀14 PCD at 45°.

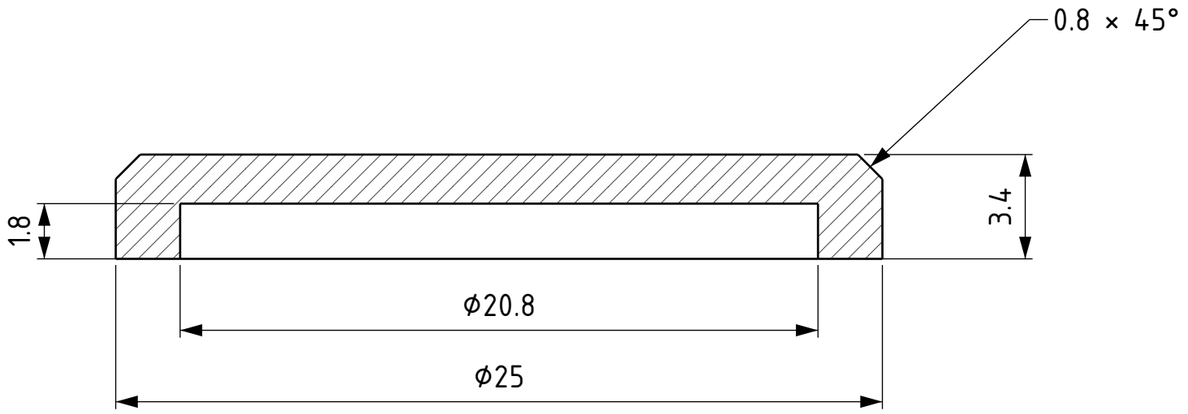
NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.
- Bolts to the jaw servo's output horn with 4x M3 x 6; closed position is jaw X along the gripper X.
- Print: horn plate down; support under the finger.

THIRD ANGLE PROJECTION

WORK SET B: forearm, wrist, gripper, moving jaw, idler plates

PROJECT TALOS 5-axis arm	EST. MASS 11 g
PART Moving jaw	SCALE 2:1
PART NO. TAL-107	UNITS / REV mm / A
QTY / ARM 1	DATE 2026-11-09
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 18 of 21



SECTION A-A SCALE 5:1

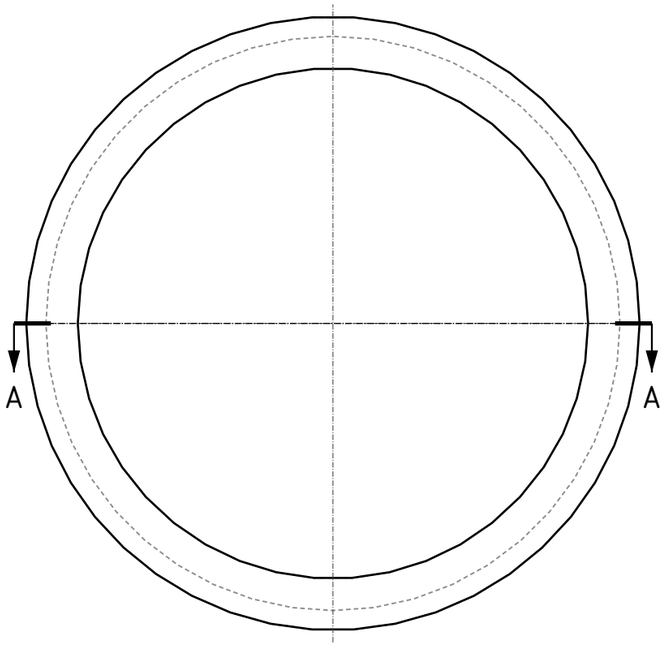
FEATURES (in order)

HUB CAP (output side, covers 4 horn screws):

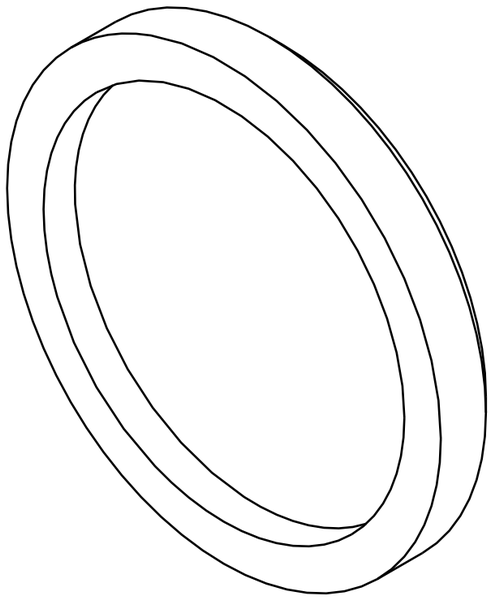
1. Disc ϕ25, 3.4 thick (y 0 to 3.4); chamfer 0.8 × 45° on the outer edge of the outer face.
2. Recess ϕ20.8, 1.8 deep from the inner face (clears M3 button heads, 1.65 high).

NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.
- Press-fit or glue onto the output-side clevis plate over the four horn screws.
- Print: outer face down; no support.



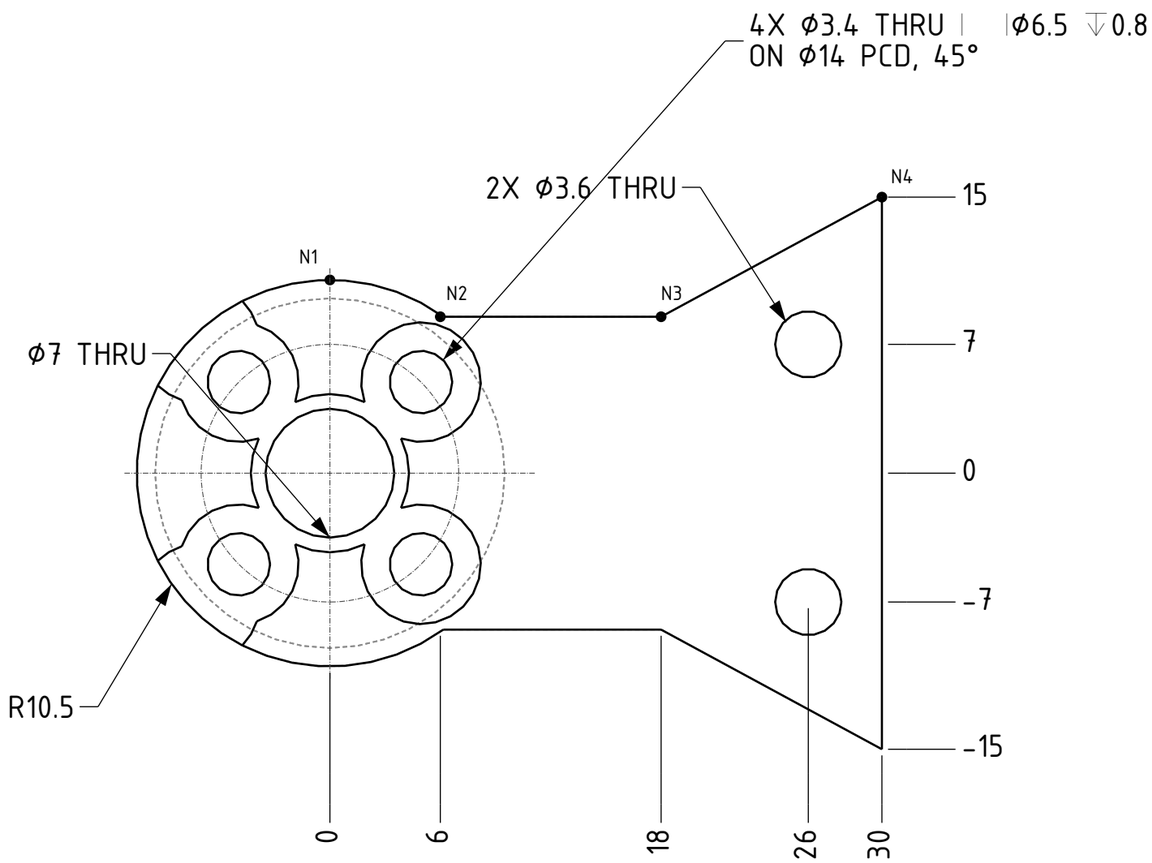
FRONT (INNER FACE)



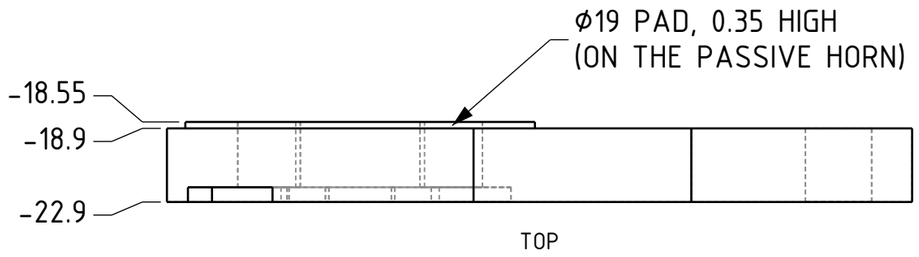
ISOMETRIC (NOT TO SCALE)

THIRD ANGLE PROJECTION

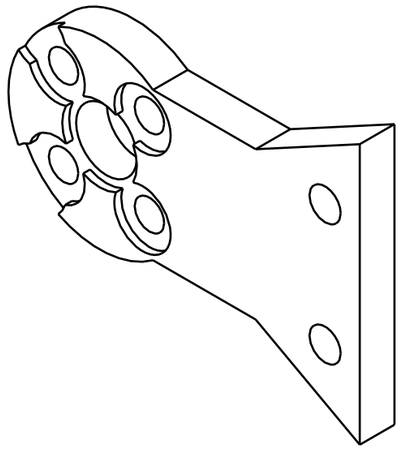
WORK SET A: base, shoulder, upper arm, hub caps, fit gauge	
PROJECT TALOS 5-axis arm	EST. MASS 1 g
PART Hub cap	SCALE 4:1
PART NO. TAL-108	UNITS / REV mm / A
QTY / ARM 3	DATE 2026-11-09
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 19 of 21



FRONT (OUTER FACE, -Y)



TOP



ISOMETRIC (NOT TO SCALE)

NECK POINTS (X, Z), MIRRORED IN Z

PT	X	Z
N1	0	±10.5
N2	6	±8.5
N3	18	±8.5
N4	30	±15

FEATURES (in order)

1. Plate, y -22.9 to -18.9: circle R10.5 at the origin + neck polygon N1 to N4 (mirrored in Z). Same outline as the idler plate on TAL-002.
2. Pad ⌀19 on the inner face, y -18.9 to -18.55: it bears on the servo's passive horn.
3. Centre hole ⌀7 through: clears the servo's idler boss (nothing screws into the boss).
4. Horn holes: 4X ⌀3.4 through on ⌀14 PCD at 45°, counterbore ⌀6.5 0.8 deep from the outer face (4X M3 × 5 into the passive horn).
5. Counterbore clean-up, 0.8 deep from the outer face: ⌀8.6 on the centre (no thin web between the counterbores and the centre hole); the two counterbores at x < 0 run out through the hub rim (second ⌀6.5 3 further out along each one's radius), so no sliver is left on the first layer.
6. Tab holes: 2X ⌀3.6 through at x 26, z ±7 (M3 × 10 into the arm link).

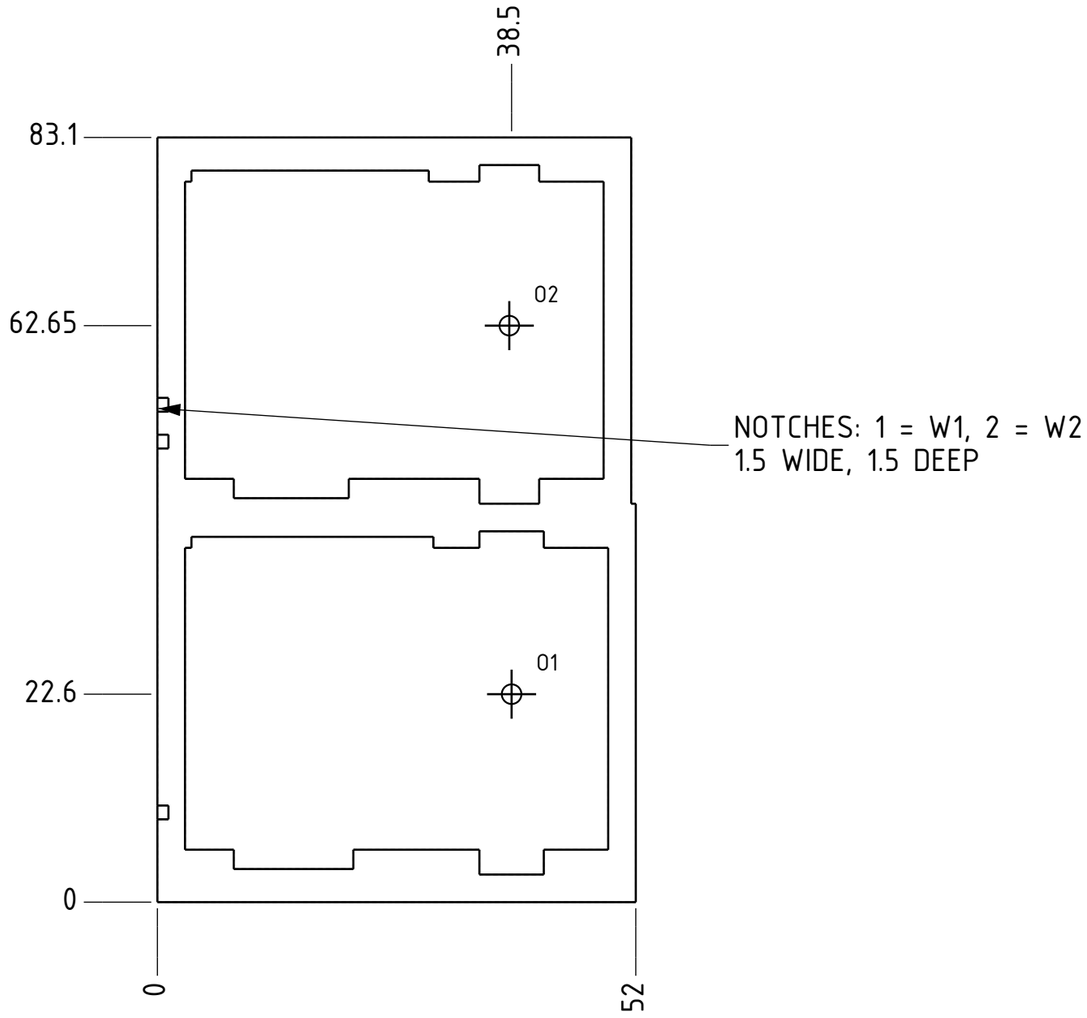
NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Part frame = the frame the part is built in, in talos_arm.py (given on each sheet).
- Printed tolerance +/-0.2 unless noted; servo pocket clearances are already included.
- One on each arm link (TAL-103, TAL-104), in the pocket on the link's -Y side: 2X M3 × 10 into the link, 4X M3 × 5 into the servo's passive horn, which turns on the idler boss (an M3 × 6 would reach the servo case behind the 1.95 thick horn). No hub cap on this side: one wide enough for the screws would hit the parent servo's plugs.
- Print: outer face down; no support.

THIRD ANGLE PROJECTION

WORK SET B: forearm, wrist, gripper, moving jaw, idler plates

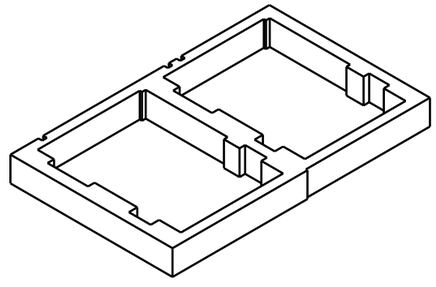
PROJECT TALOS 5-axis arm	EST. MASS 3 g
PART Idler plate	SCALE 3:1
PART NO. TAL-110	UNITS / REV mm / A
QTY / ARM 2	DATE 2026-11-09
MATERIAL PLA, 20 % infill, 3 perimeters	SHEET 20 of 21



TOP (AS PRINTED)



FRONT



ISOMETRIC (NOT TO SCALE)

GAUGE WINDOW AT ZERO CLEARANCE (X, Y)

RECTANGLE	X	Y
Case	−35.1 to 10.1	−16.5 to 15.5
Rib notch	−34.3 to −9	0 to 16.6
Spline notch	−3 to 3	0 to 17.2
Bump notch	−29.7 to −17.7	−18.5 to 0
Boss notch	−3 to 3	−19.1 to 0

FEATURES (in order)

1. Two frames, 8 thick (z 0 to 8), 3 walls, sharing the middle wall; overall 52 × 83.1.
2. Window W1 about 01 (38.5, 22.6): the case rectangle grown by 0.4 and the four notches by 0.5, the clearances every servo pocket uses. Cut through Z.
3. Window W2 about 02 (38.25, 62.65): the same, grown by 0.15 and 0.25 (tight).
4. Notches on the x 0 wall's top edge: one beside W1, two beside W2.

NOTES

- All dimensions mm. Third-angle projection. Do not scale the drawing.
- Use: take both horns off and push the bare servo (no cables) through each window along its width, top face (rib, spline) toward the rib notch, bottom face (bump, idler boss) toward the boss notch.
- It should go through W1 by hand with light friction and not through W2. If W1 binds, or W2 lets it through, tune the printer (flow, horizontal expansion) before printing the arm.
- Print: flat; no support. Print it first, with the same settings as the arm.

THIRD ANGLE PROJECTION

WORK SET A: base, shoulder, upper arm, hub caps, fit gauge

PROJECT	EST. MASS
TALOS 5-axis arm	11 g
PART	SCALE
Servo fit gauge	3:2
PART NO.	UNITS / REV
TAL-111	mm / A
QTY / ARM	DATE
1	2026-11-09
MATERIAL	SHEET
PLA, 20 % infill, 3 perimeters	21 of 21